

Forward-response supervision of a stable double-gyre Fourier neural operator

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Abstract

A Fourier neural operator (FNO) emulator of a MITGCM double-gyre configuration can forecast the ocean state well while its reverse-mode gradient bears almost no resemblance to the ocean model’s true adjoint. This report presents a matched pair of models trained to test whether that gap can be closed using forward information alone: a *control* model trained with eight-term forecast objective, and a *response* model that is identical in every respect except for one added loss term matching the emulator’s finite-difference perturbation response to MITGCM’s. No adjoint quantity, from either side, enters training or model selection. Both arms are compared forward, over a 2,000-day held rollout, and in the adjoint, against MITGCM differentiated by TAF, at three sea-surface-height targets spanning the basin, two cost functionals, and leads of 10, 30 and 90 days. The forward comparison shows the two arms follow the same evolution out to roughly two years and then separate, with the response arm drifting past climatology where the control arm stays close to it. The adjoint comparison shows the response term reliably reduces the sensitivity error — relative L_2 falls in every one of the 18 (target, objective, lead) cells reported here, by up to 32% — but does not repair its structure: pattern correlation against the true adjoint stays between -0.15 and $+0.18$ everywhere, for both arms. The one exactly known mode, the conserved basin-mean sea level, is damped by both models at a near-constant factor of 0.38 per ten-day call. Response supervision therefore corrects sensitivity magnitude without recovering sensitivity structure.

1. Introduction

MITGCM and models like it solve the discretized primitive equations directly and are accurate, but expensive: a single integration at useful resolution can take days of wall-clock time, and the ensembles that data assimilation, uncertainty quantification and sensitivity studies all require multiply that cost by however many members are needed. Data-driven emulation offers a shortcut. Rather than solving the equations, an emulator learns a single Markov map from the ocean state at time t to the state one interval later, $x(t) \mapsto x(t + \Delta t)$,

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from a library of MITGCM trajectories; once trained, that map can be applied autoregressively to produce forecasts, and ensembles of forecasts, at a marginal cost far below the original solver’s. Fourier neural operators are particularly well suited to this: FNOs naturally learn global interactions between spatial fields, whereas traditional neural networks require architectural mechanisms such as convolutions to exploit spatial structure. Bire et al. [1] trained one to emulate the fine, turbulent MITGCM double-gyre configuration autoregressively and showed it stays stable and skillful over rollouts of thousands of days, far beyond the lead times used in training.

Forecast skill is not the only property such an emulator needs, and for the applications that motivate building one in the first place it is not even the most important one. Data assimilation, parameter estimation and sensitivity analysis all depend on the *derivative* of the flow map, $\partial J/\partial x_0$, the sensitivity of some later cost functional J to the initial state — not on the map itself. This derivative, computed by the adjoint operator, reveals which earlier locations and variables a later quantity of interest actually depends on; it is the object underlying four-dimensional variational data assimilation, observing-system design, and optimal control in ocean and climate models generally. Nothing in a forecast-only training objective constrains this derivative directly, only the states it happens to be evaluated on, so a model can reproduce trajectories well while its derivative is wrong. This is not a hypothetical concern. Ruiz [2] built a reverse-mode adjoint pipeline for Samudra [3], a fast neural ocean emulator with genuinely strong forward skill, and compared it directly against MITGCM’s: Samudra’s sensitivities were internally self-consistent but diffused far faster than MITGCM’s, alternated sign on adjacent grid cells where MITGCM’s stayed spatially coherent, and showed no trace of the westward-propagating Rossby-wave signal that MITGCM’s adjoint carries clearly at long lag. A follow-on internship at JPL of Mousavi [4], motivated explicitly by that result, abandoned differentiating the forward emulator altogether and instead trained a separate network to emulate the adjoint recursion itself — a tacit acknowledgment that a good forward emulator’s own derivative could not be trusted directly.

JPL’s specific advantage in pursuing this question further is access to a mature, TAF-differentiated MITGCM adjoint, verified against finite differences and used routinely for ocean state estimation. That reference adjoint makes a direct, apples-to-apples comparison possible: differentiate a trained emulator in reverse mode, differentiate MITGCM through TAF over the identical trajectory and cost functional, and compare the two sensitivity maps cell by cell rather than relying on qualitative plausibility checks alone.

The scientific question this line of work poses is therefore simple to state: does a good forward emulator also learn the correct sensitivities? On the evidence assembled so far — Samudra against MITGCM, the dedicated adjoint surrogate’s own limitations at long lag, and this project’s own production FNO — the answer is that it is not guaranteed. That diagnosis suggests a targeted remedy. A Jacobian-vector product is, to leading order, a forward finite difference; a large enough set of MITGCM perturbation experiments therefore constrains the emulator’s Jacobian using only forward integrations, with no adjoint code, no TAF output, and no risk of training on the very quantity the study is meant to test. This report presents the two models that test that idea, both trained from random initialization

on the weakly turbulent S0 double gyre:

- the **control model** (arm B, `model_c_adjoint_faithful_nominal_control_v1`), trained with the project’s existing eight-term forward objective and nothing else;
- the **response model** (arm C, `model_c_adjoint_faithful_response_v1`), identical in architecture, data, split, optimizer, schedule and objective, plus exactly one additional loss term that matches signed forward perturbation responses against MITGCM.

Section 2 describes the two models and the design choices behind them. Section 3 states the evaluation protocol. Section 4 reports the forward comparison and then the adjoint comparison at three targets and two cost functionals. Sections 5 and 6 interpret the outcome.

2. Methodology

2.1. Problem formulation

The emulator is a single Markov operator advancing the full prognostic ocean state ten days,

$$F_\theta : [x_t, S] \mapsto x_{t+10},$$

on the MITGCM `tutorial_baroclinic_gyre` configuration at one degree (62×62 cells, 15° – 75° N), deliberately coarser and less turbulent than the 0.25° configuration of Bire et al. [1]: in a weakly turbulent regime trajectories do not diverge chaotically (Section 3), so a mismatch between the emulator’s and MITGCM’s Jacobians can be attributed to the emulator itself rather than to sensitive dependence on initial conditions, making this a cleaner first testbed before attempting the same comparison in Bire et al.’s fully turbulent regime. The state $x \in \mathbb{R}^{46 \times 62 \times 62}$ holds fifteen vertical levels each of zonal velocity U , meridional velocity V and potential temperature Θ , plus the free surface η ; S holds five static physical fields (wind stress, wet mask, Coriolis parameter, grid spacing, SST relaxation target) and a two-channel position encoding. Keeping the state closed — every prognostic variable, no diagnostic reductions — is what makes the Jacobian of F_θ a candidate tangent-linear operator at all; a reduced state would not support one. Training and inference are done in pointwise-normalized coordinates,

$$\mathcal{N}(x)_c(y, x) = \frac{x_c(y, x) - \mu_c(y, x)}{\sigma_c(y, x)},$$

with μ, σ computed from training days only and land set to zero. A forecast to lead $T = 10k$ days is k self-fed applications of F_θ .

Writing M_k for the true MITGCM flow map over k ten-day steps and $F_{\theta,k}$ for the emulator’s, forecast skill is a statement about $F_{\theta,k}(x) \approx M_k(x)$, whereas adjoint fidelity is a statement about their derivatives, $DF_{\theta,k}^\top \approx DM_k^\top$. The object we compare is the sensitivity map of a scalar cost J_T with respect to the initial sea surface,

$$S_T[j, i] = \frac{\partial J_T}{\partial \eta(j, i, t_0)}, \quad (1)$$

a dimensionless influence coefficient: the fraction of a unit sea-level perturbation at cell (j, i) that survives in J after T days. It is exactly zero on the 244 land cells, where sea-surface height is not a degree of freedom, and is spread over the $\sim 3,600$ wet cells, so individual values of order 10^{-4} are the physically expected scale, not a weak signal.

2.2. Architecture and design choices from Bire et al.

The overall map, state convention and Fourier-operator backbone follow the FNO ocean-emulation architecture of Bire et al. [1], who developed and validated this class of emulator for the double gyre at $0.25^\circ \times 0.25^\circ$; the present models adapt it to a one-input Markov map at one degree. The network is a three-block Fourier neural operator with 32×32 retained Fourier modes and hidden width 128, a parallel bias-free 3×3 local correction branch and six pointwise channel LayerNorms, for 27,297,960 parameters (Table 1). The rationale behind the main choices is short:

- **Global spectral convolutions.** Barotropic adjustment in this basin is fast and non-local: a sea-level perturbation anywhere influences the whole domain within days. A convolutional stencil cannot represent that in one ten-day step; a Fourier layer can, because each retained mode couples the entire grid at once.
- **A local 3×3 branch alongside it.** The western boundary current is a few grid cells wide and is exactly what a truncated spectral basis represents worst. The local branch is bias-free and zero-initialized, so it starts as an exact no-op and only earns weight where the spectral branch is insufficient.
- **Direct state prediction, not residual.** The map outputs x_{t+10} itself. This keeps the recurrence a genuine flow map, so that composing it k times and differentiating gives a well-defined k -step tangent operator.
- **Per-mode spectral normalization.** Inside each spectral convolution, the dense complex 128×128 channel-mixing matrix at each of the 1,632 retained modes is capped to unit spectral norm by two power iterations per forward pass [5]. The cap is one-sided: a mode already below the cap is left untouched and carries no gradient. It bounds one constituent of each block rather than the full block Jacobian, so legitimate transient amplification through the skip connections, channel MLP and local branch remains representable. This is the project’s answer to the autoregressive blow-up that unconstrained arms exhibited over 2,000-day rollouts, and it was chosen over alternatives — mode truncation, removing the local branch, a saturating tanh stabilizer — precisely because those buy stability by suppressing structure or by manufacturing artificially small derivatives, which is the wrong trade for an emulator whose derivative is the object of study.

Item	Value
Grid / interval	62×62 , $\Delta t = 10$ days, direct state (not residual)
Fourier modes	32×32
Hidden width	128
FNO blocks	3
Local branch	bias-free 3×3 , zero-initialized
Spectral normalization	per-mode $\sigma_{\max} \leq 1$, 1,632 modes
Parameters	27,297,960 (97.95% in the spectral convolutions)

Table 1: Core architecture, byte-for-byte identical between the control and response arms.

2.3. Control model

The control model is trained from random initialization with the project’s eight-term, fully dimensionless forward objective,

$$L_{\text{nominal}} = L_{\text{state}} + 0.001 L_{\text{inc}} + 0.50 L_{\text{rollout}} + 10^{-5} L_{\text{spec}} + 0.065 L_{\text{bdy}} + 0.05 L_{\text{pg}} + 0.05 L_{\text{cont}} + 0.05 L_{\text{bt}}, \quad (2)$$

whose terms penalize, in order, the state error, the ten-day increment, a six-call autoregressive rollout, the radial power spectrum, the four-cell western boundary band, the pressure-gradient balance, continuity, and barotropic transport. The four physics terms add no output channels; they are diagnostic quantities computed from the predicted state and compared against the same quantities computed from truth. All eight terms are active from step 1.

Training is 7,680 Adam steps at an effective batch of eight 60-day sequences (realized as two microbatches of four), $\beta = (0.9, 0.95)$, learning rate 5×10^{-4} dropping to 10^{-4} for the last quarter, no teacher forcing after the initial state — 368,640 autoregressive state transitions, 3.38 h on one V100. Checkpoints at steps 1,920/3,840/5,760/7,680 are each autoregressed 360 days from 102 pooled validation starts and selected by a three-part rule: short-horizon (10–90 d) skill within 5% of the best candidate in every primary field; a measured twin-perturbation growth rate at or below 1 per call; and, among survivors, the smallest 90–360 d error relative to a train-only climatology. No checkpoint met the growth-rate ceiling, so the declared fallback applied and step 7,680 was published ($\hat{\lambda} = 1.0126$ per call, worst long-horizon ratio 0.288).

2.4. Response model

The response model is the control model plus one loss term. Its motivation is the central-difference identity: for a direction v and amplitude ϵ ,

$$\frac{M_k(x + \epsilon v) - M_k(x - \epsilon v)}{2\epsilon} = DM_k(x) v + O(\epsilon^2), \quad (3)$$

so matching the emulator’s forward perturbation response to MITGCM’s constrains Jacobian-vector products along the sampled directions — using nothing but ordinary forward integrations. By duality, for any cotangent w ,

$$\langle (DF_k^T - DM_k^T) w, v \rangle = \langle w, (DF_k - DM_k) v \rangle, \quad (4)$$

so a better-matched forward response implies a better-matched *adjoint projection* onto the sampled direction span — though not the full Jacobian in unsampled directions, which is exactly what the blind adjoint test is designed to probe.

Concretely, at each sampled anchor state x_q , input group $h(q) \in \{U, V, \Theta, \text{SSH}\}$, direction v_q and signed amplitude $s\epsilon_q$ with $s \in \{-1, +1\}$, MITGCM is integrated from $x_q + s\epsilon_q v_q$ and from the unperturbed x_q , and the normalized signed response is recorded at every ten-day lead k ,

$$r_{M,q,k}^s = \frac{\mathcal{N}[M_k(x_q + s\epsilon_q v_q)] - \mathcal{N}[M_k(x_q)]}{s\epsilon_q}, \quad (5)$$

with $r_{F,q,k}^s$ the same quantity computed from the emulator on the same edited initial states. Writing $g \in \{U, V, \Theta, \text{SSH}\}$ for an output group and $d_{h,g,k}^2$ for the training-data mean-square response of output group g to a perturbation of input group h at lead k , the added term is

$$L_{\text{response}} = \text{mean}_q \frac{1}{8} \sum_{s \in \{-1, +1\}} \sum_g \frac{\text{mean}_{c \in g, \text{wet}} (r_{F,q,k}^s - r_{M,q,k}^s)^2}{d_{h(q),g,k}^2}, \quad (6)$$

averaged over sampled directions q and, for directions carried past ten days, over lead k as well. The normalization by $d_{h,g,k}^2$ puts every input/output pair on a comparable unitless scale, so no group dominates simply by having larger physical units or more vertical levels. The total objective is

$$L_{\text{total}} = L_{\text{nominal}} + \lambda_{\text{resp}} L_{\text{response}}, \quad \lambda_{\text{resp}} = 10^{-3}. \quad (7)$$

Three design choices are worth naming. First, the response is *signed and oriented*: both $s = +1$ and $s = -1$ are supervised separately rather than being combined into a symmetric difference, so the model is asked to reproduce the direction of the response, not only its magnitude. Second, *no ordinary state loss is placed on the perturbed trajectories* — the signed response is the only quantity supervised on them, so the term cannot degenerate into extra forecast training. Third, the response term fires on a predeclared subset of steps (every fourth optimizer step, 1,920 auxiliary updates, cycling `short, short, short, long` over the lead-10 and the lead-10–60 pools), and the auxiliary forward chain snapshots and restores the spectral-normalization power vectors bit-for-bit, so it leaves no persistent state behind and the control arm’s numerics are reproduced exactly when the term is switched off. Nothing else differs: same seed, same data, same split, same schedule, same eight terms, same selection rule. Training took 3.08 h on one V100, and the same fallback branch published step 7,680 ($\hat{\lambda} = 1.0138$ per call, worst long-horizon ratio 0.316).

Critically, **no adjoint quantity enters anywhere** — not MITGCM/TAF output, not the emulator’s own reverse-mode gradient — in generating the response data, in Eq. (6), in choosing λ_{resp} , or in selecting a checkpoint. Every number in Section 4.2 is therefore a genuinely out-of-sample test.

Seed sensitivity. These models are strongly sensitive to the random seed: across the study’s three paired seeds the control arm’s day-2,000 surface pressure error ranges from 0.22 to 5.79

times climatology, and the seed that gives the best control model is not the seed that gives the best response model. We present seed 20260911 throughout because it produced the best-forecasting control model of the three, judged on forward evidence alone; the study’s own frozen contract selects no best seed and reports all three, and the confirmatory verdict quoted in Section 5 remains that of the pre-declared primary seed.

3. Experimental setup

Regime and data. All results use S0, the control wind amplitude ($\tau_0 = 0.100 \text{ N m}^{-2}$) matching Bire et al.’s [1] original configuration. Training uses days 0–6,000 (5,940 starts per regime, 17,820 sequences pooled over the three wind regimes S0/S1/S2), validation days 6,000–7,200, and held inference 15 fixed S0 starts never used for training or selection, autoregressed to day 2,000 against lead-matched MITGCM truth. S0 is weakly turbulent by design: twin MITGCM pickups differing by a 10^{-3} relative perturbation do not separate over 25 simulated years, so a long-rollout discrepancy between emulator and truth is model error, not chaotic divergence being amplified — the property that makes a Jacobian comparison interpretable.

Response dataset. Perturbation amplitudes were fixed by a forward-only pilot before any response training, one per input family (U and V at $\alpha = 0.1$, Θ at 0.005, SSH at 0.05, in units of the family’s local variability), chosen as the largest amplitude at which the response still behaves linearly and stays above the solver’s own numerical floor. The training split holds 672 sampled directions (576 supervised at lead 10, 96 carried to lead 60) balanced across input families, regimes and both signs, with a separate validation split used only for reporting and a blind split ($\sim 10,890$ MITGCM model-days over 441 branches) held out entirely.

Adjoint comparison. Sensitivity maps, Eq. (1), are produced two ways on the identical S0 trajectory and window: through MITGCM differentiated by TAF 6.8.11 (reference), and by reverse-mode automatic differentiation of the trained emulator (test). The emulator variant reported throughout is S_{forced} , which linearizes the emulator about the *true* MITGCM trajectory at every ten-day step, so it isolates Jacobian error from trajectory drift; the free-running variant S_{free} , which linearizes about the emulator’s own rollout (Appendix Appendix A.1 derives exactly how the two relate), agrees with it to within a fraction of a percent for both arms here, so the two are not shown separately. Both producers pass their own verification independently — the TAF gradient against direct finite differences of the forward model, exactly zero at every land cell and reproducing the analytically known basin-mean conservation; the emulator’s reverse-mode gradient against forward-mode differentiation, against finite differences of the trained operator, and against the adjoint identity $\langle v, Ju \rangle = \langle J^T v, u \rangle$ to near machine precision. Appendix Appendix A details how each side is produced and checked, file by file, for a reader who did not write either pipeline.

Three target cells are used, all on the same latitude row $j = 17$ so that the only variable is distance from the western wall: **western** $(i, j) = (2, 17)$, at the boundary current; **interior** $(31, 17)$; and **eastern** $(61, 17)$. All three were named and gradient-checked in the frozen

MITGCM adjoint record before either model finished training. Two cost functionals are reported:

$$J_{\text{raw}} = \eta(p^*, T), \quad (8)$$

$$J_{\text{anom}} = \eta(p^*, T) - \langle \eta(T) \rangle_A, \quad \langle \eta \rangle_A = \sum_{j,i} w_{ji} \eta_{ji}, \quad \sum_{j,i} w_{ji} = 1, \quad (9)$$

where w is the cell-area weight ($w_{ji} \propto \cos \phi_j$), plus the basin-mean functional $J_{\text{mean}} = \langle \eta(T) \rangle_A$ on its own. Since $J_{\text{raw}} = J_{\text{anom}} + J_{\text{mean}}$ and all three are linear in $\eta(T)$ for a fixed trajectory, the three reference maps must satisfy $S^{\text{raw}} = S^{\text{anom}} + S^{\text{mean}}$ exactly; this holds in the extracted MITGCM products to a worst relative L_2 of 4.05×10^{-8} , an independent confirmation that the weight fields are wired correctly. Leads are $T \in \{10, 30, 90\}$ days.

Metrics. Four scalars compare an emulator map S^F against the reference S^M over wet cells only:

$$\rho = \text{corr}_{\text{wet}}(S^F, S^M) \quad (\text{pattern correlation: is it in the right places?}) \quad (10)$$

$$\varepsilon_2 = \frac{\|S^F - S^M\|_{2,\text{wet}}}{\|S^M\|_{2,\text{wet}}} \quad (\text{relative } L_2: \text{total error}) \quad (11)$$

$$\alpha = \frac{\|S^F\|_{2,\text{wet}}}{\|S^M\|_{2,\text{wet}}} \quad (\text{amplitude ratio: over- or under-response}) \quad (12)$$

$$\sigma = \text{mean}_{\text{wet}} \mathbf{1}[\text{sign } S^F = \text{sign } S^M] \quad (\text{sign agreement}) \quad (13)$$

Pattern correlation is scale-free, so a map with the right structure and the wrong amplitude still scores 1; the amplitude ratio is its companion. Note the consequence of $\rho \approx 0$: with no structural agreement, ε_2 and α become nearly equal, since $\|S^F - S^M\|^2 = \|S^F\|^2 + \|S^M\|^2 - 2\langle S^F, S^M \rangle$ and the cross term vanishes. Every table below shows this. Sign agreement near 0.5 is what an uncorrelated map gives.

4. Results

4.1. Control-response comparison, forward

Table 2 collects the forward numbers for the two arms. The two models are close at short lead and separate at long lead, in the control's favour.

Metric (S0, 15-member held inference)	Control	Response	Truth / target
Day-200 RMSE, surface P/ρ ($\text{m}^2 \text{s}^{-2}$)	0.0054	0.0120	—
Day-200 RMSE, SST ($^{\circ}\text{C}$)	0.0132	0.0135	—
Day-200 RMSE, surface speed (m s^{-1})	0.00139	0.00139	—
Day-200 ACC, surface P/ρ	0.9968	0.9798	1
Day-200 ACC, SST	0.9466	0.9442	1
Day-2,000 RMSE / climatology, surface P/ρ	0.222	1.501	< 1
Day-2,000 RMSE / climatology, SST	0.633	1.779	< 1
Day-2,000 RMSE / climatology, surface speed	1.210	2.345	< 1
Max. normalized amplitude through day 2,000	6.55	9.89	≤ 8
Twin-perturbation growth rate per call	1.0126	1.0138	≤ 1
Day-2,000 ψ' RMS (Sv)	0.81	1.67	0.74
Day-2,000 western/interior ψ' ratio	6.23	7.23	23.1

Table 2: Forward comparison, seed 20260911, S0, 15 held starts never used for training or selection. Bold marks the better arm. The control arm is better on every metric except the western-boundary concentration ratio, where both remain far below truth.

Error growth to day 2,000 (Figure 1). This is the project’s figure-8 diagnostic: pooled 15-member RMSE against lead-matched MITGCM truth for surface speed, surface P/ρ and SST, from day 0 to day 2,000, with two reference curves. *Persistence* holds the initial state fixed and so measures how much the ocean itself changes over the lead; *climatology* predicts the training-mean state and so measures the error of knowing nothing but the mean seasonal cycle. A useful emulator must stay below both. The two arms present the same evolution to roughly day 700. After that the response arm (purple) rises steadily and crosses climatology near day 1,500 in SST and near day 1,900 in surface P/ρ , while the control arm (red) stays below climatology to day 2,000 in both. The shaded interdecile bands are the more telling difference: the response arm’s day-2,000 surface-pressure band runs from 0.029 to 0.362, a spread of more than an order of magnitude across members, against 0.017–0.020 for the control. The failure is therefore not a uniform drift but a few members losing control late in the rollout — consistent with its higher worst-case normalized amplitude (9.89 against 6.55, both compared against a declared ceiling of 8) and its slightly larger self-amplification rate.

Barotropic streamfunction (Figures 2 and 3). The first figure shows the total barotropic streamfunction ψ for truth, control and response at day 60 and day 2,000; the second subtracts the MITGCM S0 training-block time mean, leaving the transient anomaly $\psi' = \psi - \bar{\psi}_{\text{S0}}$, the project’s figure-7a diagnostic. Like sea surface height, the total field is a weak test: both arms reproduce the subpolar and subtropical gyres — their strength, sign and western intensification — essentially indistinguishably from truth at both day 60 and day 2,000, because the time-mean circulation dominates the field and reproducing a mean is easy. The anomaly view is where the arms separate. Truth’s day-2,000 transient anomaly is sharply localized: a small dipole in the western boundary current near 31°N and almost nothing elsewhere, with RMS 0.74 Sv and a western-band-to-interior concentration ratio

of 23.1. The control reproduces the localization and its amplitude well (0.81 Sv, ratio 6.23) while spreading the anomaly too broadly. The response arm instead develops a strong coherent western band extending the full meridional extent of the basin plus a broad negative interior — 1.67 Sv, 2.2 times truth’s amplitude. Its concentration ratio is nominally the better of the two (7.23 against 6.23), which illustrates why that ratio is a poor standalone score: it improves here because the spurious anomaly happens to sit at the boundary, not because the boundary current is better resolved.

Sea surface height (Figures 4 and 5). The first figure shows the total field η for truth, control and response at day 60 and day 2,000, with truth-minus-model differences on their own scale; the second subtracts the MITGCM S0 training-block time mean (6,000 days) from all three panels alike, leaving the anomaly η' . The total-field view shows both arms reproducing the double-gyre sea-level structure — the subtropical high, the subpolar low, the tight western gradient — essentially perfectly at day 60 and still recognizably at day 2,000, with errors two orders of magnitude below the field itself. This is why the total field is a weak test: the time-mean circulation dominates it, and reproducing a mean is easy. The anomaly view removes that pedestal and is where the arms separate. At day 60 both anomalies match truth closely. At day 2,000 the control anomaly is a slightly smoothed version of truth with the right sign structure, while the response anomaly has developed a spurious basin-scale dipole — a positive band along the northern rim and across the western boundary, a negative interior pool centred near 50°N — that is absent from truth. The truth-minus-response panel is correspondingly larger and more organized than truth-minus-control.

Sea surface temperature anomaly (Figure 6). SST’ at day 60 and day 2,000. Truth’s anomaly is confined to the western boundary near 31°N, where the separated jet advects the sharp meridional temperature front. Both arms capture that feature at day 60 with the correct sign and near-correct amplitude. By day 2,000 the control has kept the boundary signal and added only mild interior noise; the response arm has developed a coherent warm plume extending east-north-east from the boundary along the jet axis, plus a spurious warm band along the southern rim. This is the spatial signature behind the response arm’s day-2,000 SST error of 1.78 times climatology against the control’s 0.63.

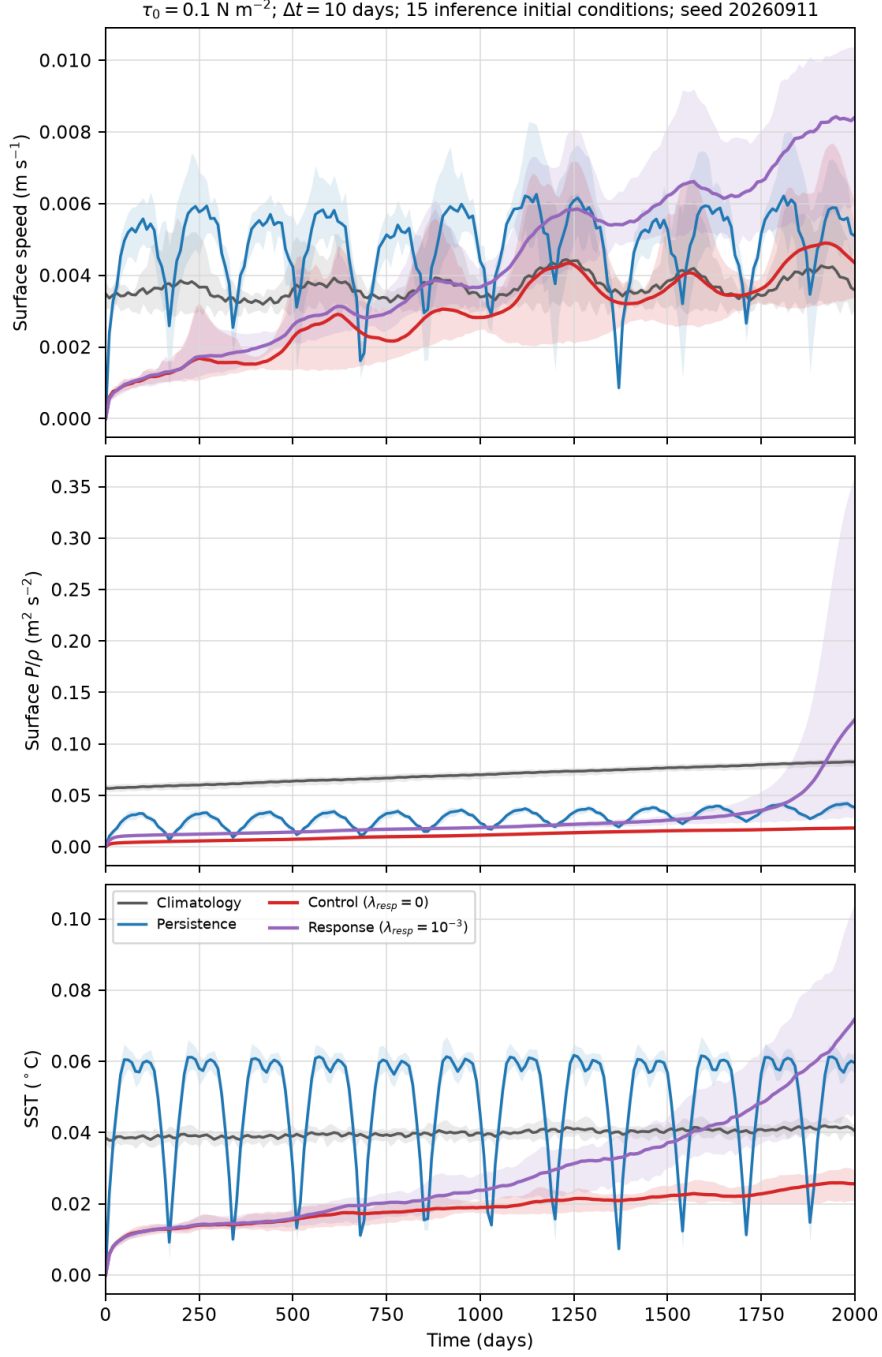


Figure 1: Pooled 15-member held-inference RMSE against MITGCM truth, $\Delta t = 10 \text{ d}$, days 0–2,000, S0, seed 20260911. Shading is the interdecile range across members. The arms are indistinguishable to $\sim$ day 800; the response arm then crosses climatology in SST and surface pressure while the control arm does not.

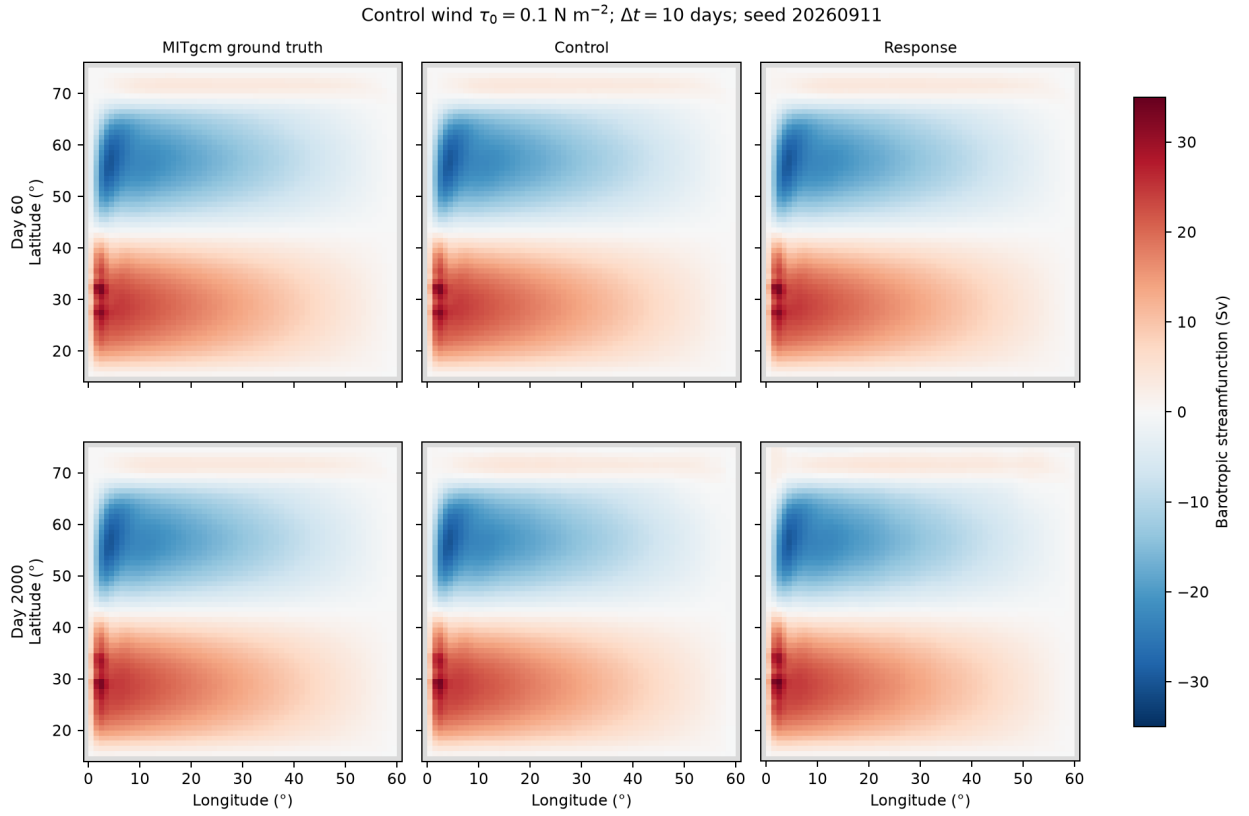


Figure 2: Barotropic streamfunction ψ , day 60 (top) and day 2,000 (bottom): truth, control and response. Both arms reproduce the total gyre circulation essentially indistinguishably from truth at both leads; Figure 3 removes the time-mean pedestal to show where they actually differ.

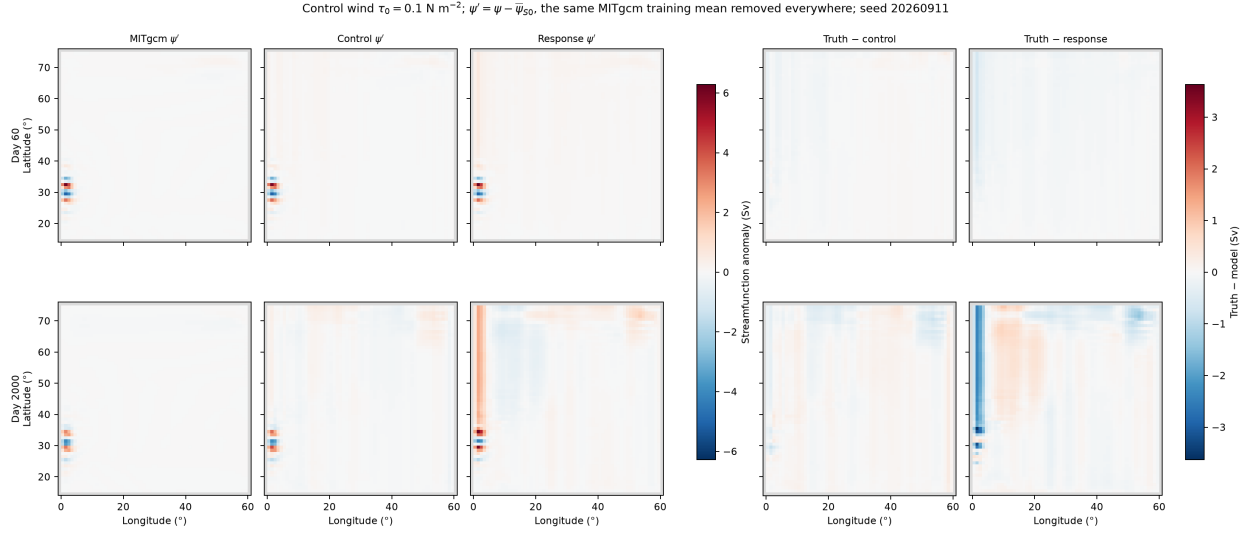


Figure 3: Barotropic streamfunction anomaly ψ' about the MITGCM S0 training mean, day 60 and day 2,000. Truth's day-2,000 anomaly is a compact western-boundary dipole (0.74 Sv RMS); the control reproduces its amplitude (0.81 Sv) and the response arm overshoots it by $2.2\times$ (1.67 Sv) with a spurious full-basin western band.

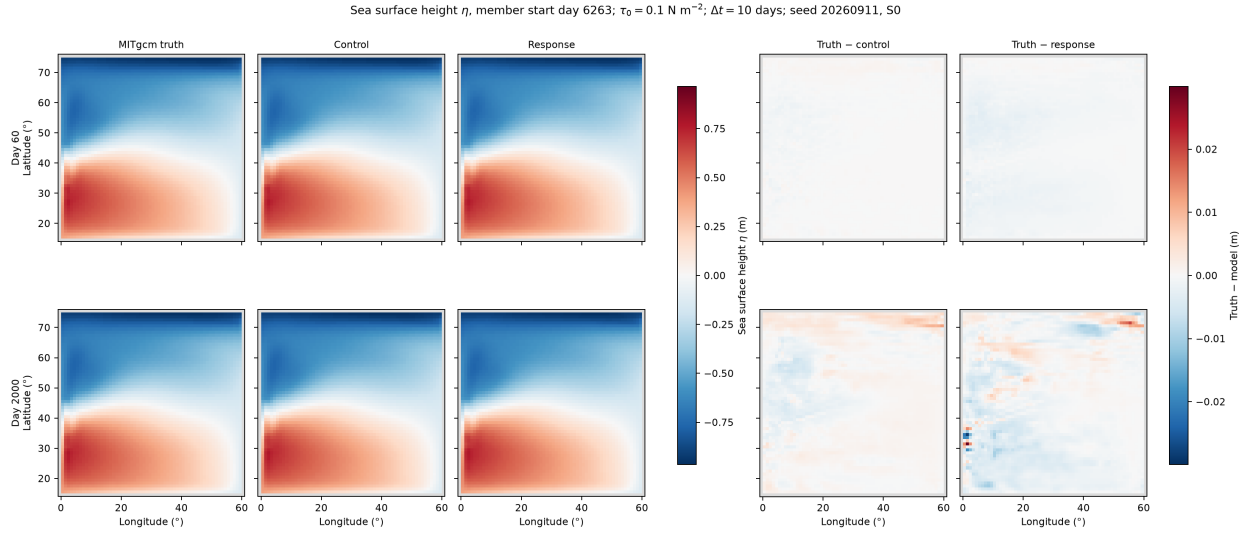


Figure 4: Sea surface height η , day 60 (top) and day 2,000 (bottom): truth, control, response, and the two truth-minus-model differences on a shared scale 40 times finer. Both arms reproduce the total field; the difference panels carry the whole story, and even at day 2,000 they are two orders of magnitude below the field.

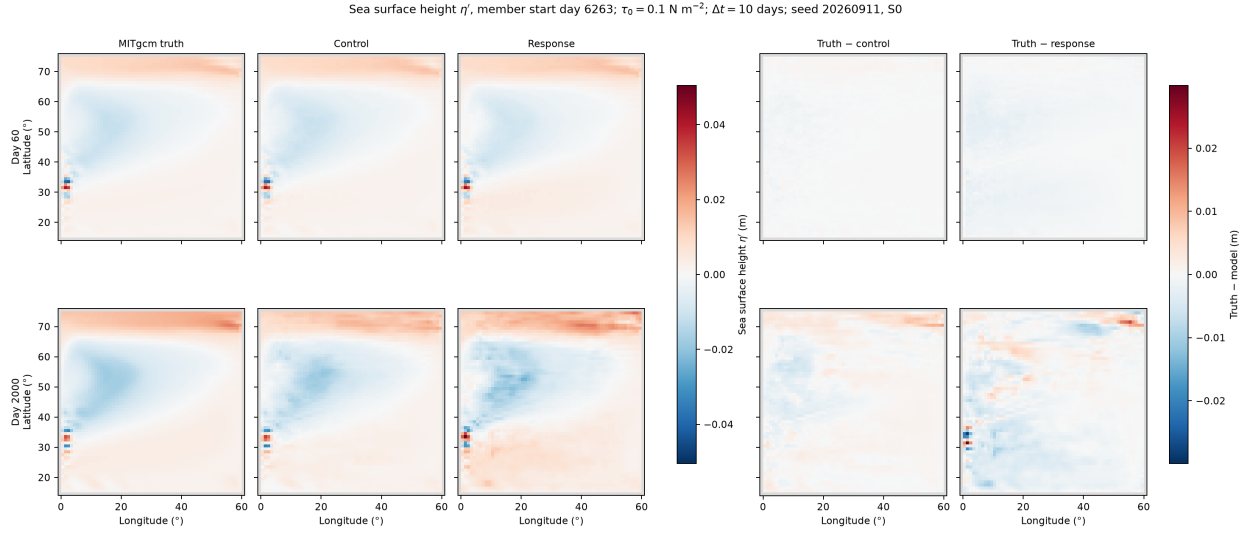


Figure 5: Sea surface height anomaly η' , the same layout with the MITGCM S0 training-block time mean removed from truth and both models alike. At day 2,000 the response arm has grown a basin-scale dipole — positive northern rim and western boundary, negative interior — that is absent from truth and from the control.

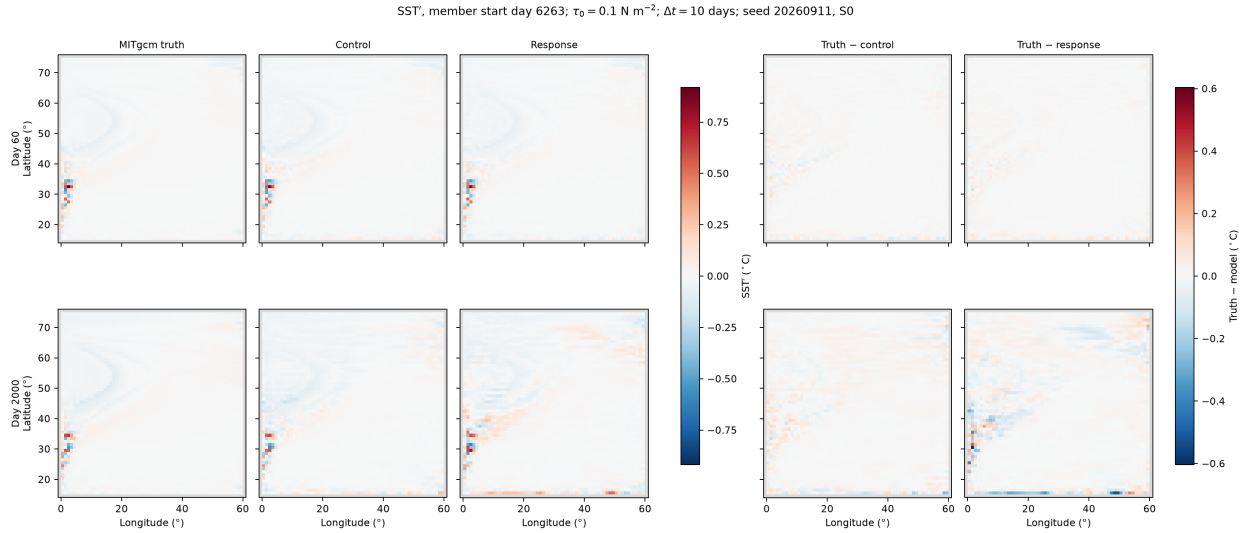


Figure 6: SST anomaly, day 60 and day 2,000. Both arms capture the western-boundary front at day 60; by day 2,000 the response arm has added a spurious warm plume along the jet axis and a warm band on the southern rim.

4.2. Control–response comparison, adjoint

Every figure in this section uses the same layout: two rows, control (`fno_b_seed_20260911`) above and response (`fno_c_seed_20260911`) below, and three panels per row — the MITGCM/TAF reference S_T , the emulator’s S_{forced} , and their difference. Columns 1 and 2 share one colour scale so the amplitudes are directly comparable; column 3 has its own. The open circle marks the target cell p^* . The four metrics of Eqs. (10)–(13) are printed above each row.

Because the two rows plot the same physical reference, everything that differs between them is attributable to the one training term.

4.2.a. Mean conservation

This is the one case whose exact answer is known analytically. With an implicit free surface and `exactConserv`, the configuration conserves the area-weighted basin-mean sea level exactly, so $J_{\text{mean}}(T) = \langle \eta(T) \rangle_A = \langle \eta(t_0) \rangle_A$ for every T , and therefore

$$S_T^{\text{mean}}[j, i] = \frac{\partial J_{\text{mean}}}{\partial \eta(j, i, t_0)} = w_{ji} \quad \text{for all } T, \quad (14)$$

independent of lead. The MITGCM/TAF panel in Figures 7–9 is therefore literally the cell-area weight field: smooth, single-signed, and $\propto \cos \phi$, hence largest in magnitude along the southern boundary where cells are widest ($\max |S| = 3.96 \times 10^{-4}$, exactly $1.43 \times$ the uniform value $1/3600$, as $\cos(15.5^\circ)$ over the domain-mean cosine requires). It is identical at every lead, verified to 3.57×10^{-8} over 91 daily dumps. The correct amplitude ratio is exactly $\alpha = 1$ at every lead, and any departure is unambiguously emulator error.

Lead (d)	Control			Response		
	ρ	ε_2	α	ρ	ε_2	α
10	+0.907	0.626	0.380	+0.927	0.632	0.373
30	+0.625	0.949	0.055	+0.698	0.950	0.053
90	+0.012	1.002	0.007	−0.031	1.000	0.003

Table 3: Basin-mean conservation. The exact answer is $\rho = 1$, $\varepsilon_2 = 0$, $\alpha = 1$ at every lead. Both arms damp the conserved mode at a near-constant factor per ten-day call.

Both models fail this test, and they fail it in a specific, interpretable way. After a single ten-day call the emulator has retained only 38% of the conserved mode ($\alpha = 0.380$ control, 0.373 response) — but with pattern correlation 0.91–0.93, meaning it has the shape of the weight field almost right and simply scaled it down. The decay is close to geometric in the number of calls: three calls give $0.380^3 = 0.0549$ against a measured 0.055, and $0.373^3 = 0.0519$ against a measured 0.053. In other words, both emulators apply an approximately constant damping factor of ≈ 0.38 per call to the one mode the ocean model conserves exactly. By lead 90 the mode is gone (amplitude ratios 0.007 and 0.003; the residual decays more slowly than the pure geometric law would predict, so a faint remnant survives), the

pattern correlation has collapsed to zero, and $\varepsilon_2 \rightarrow 1.00$ — the value ε_2 takes when the emulator map is simply absent, since $\|S^F - S^M\| \rightarrow \|S^M\|$.

This is a qualitatively different failure from every other case in this section. Everywhere else the emulator is *too* sensitive by factors of 5 to 47; here it is not sensitive enough by a factor of 3 to 300. A neutral mode — one whose true amplification factor is exactly 1 — is precisely what an autoregressive-stability regularizer suppresses, and the per-mode spectral normalization caps every channel-mixing matrix at unit norm, which cannot amplify but readily damps. Improved mean-mode preservation was a named secondary endpoint of the study; the response term moves it slightly in the wrong direction.

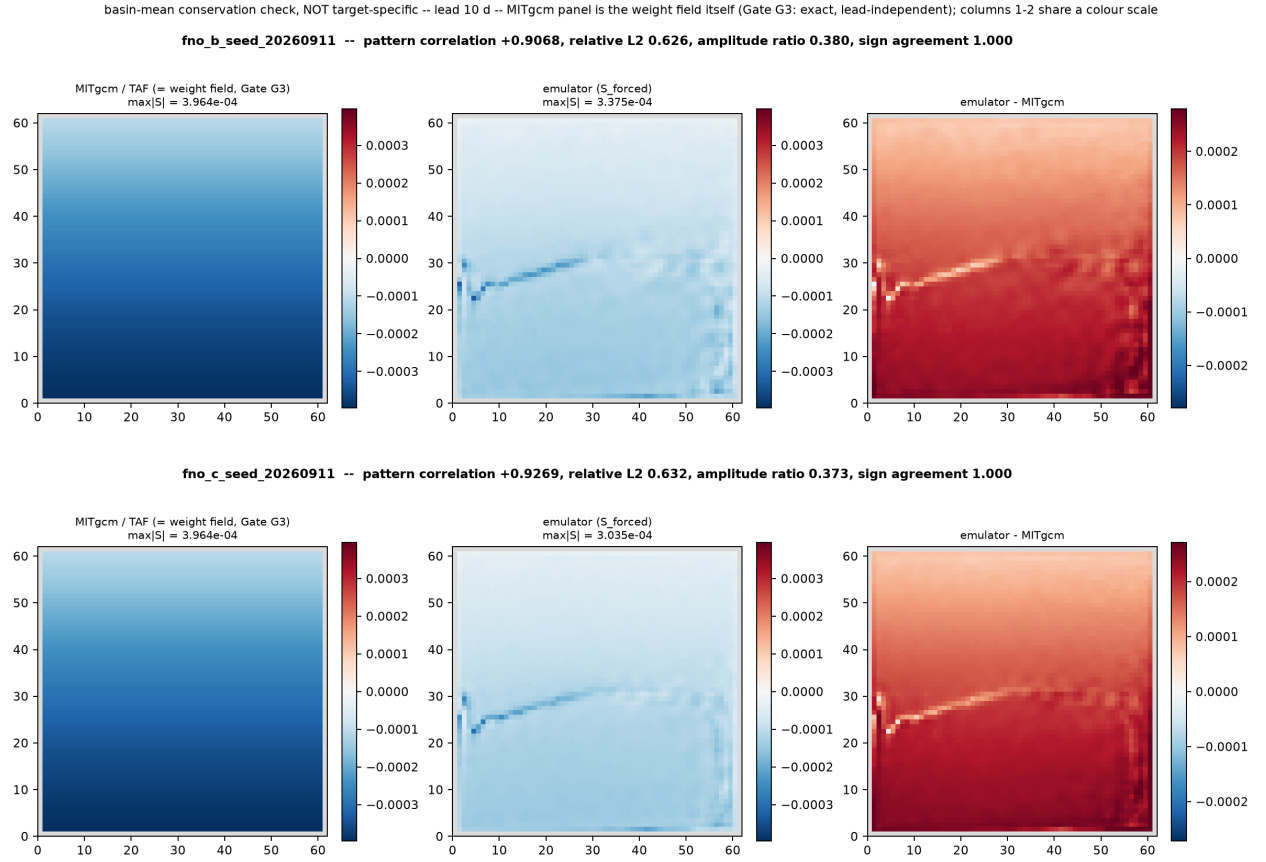


Figure 7: Basin-mean conservation, lead 10 d. The MITGCM panel is the exact area-weight field, Eq. (14). Both emulators recover its shape ($\rho \approx 0.91$ – 0.93) at 38% of its amplitude; the visible fine structure along the jet axis in the emulator panels is spurious.

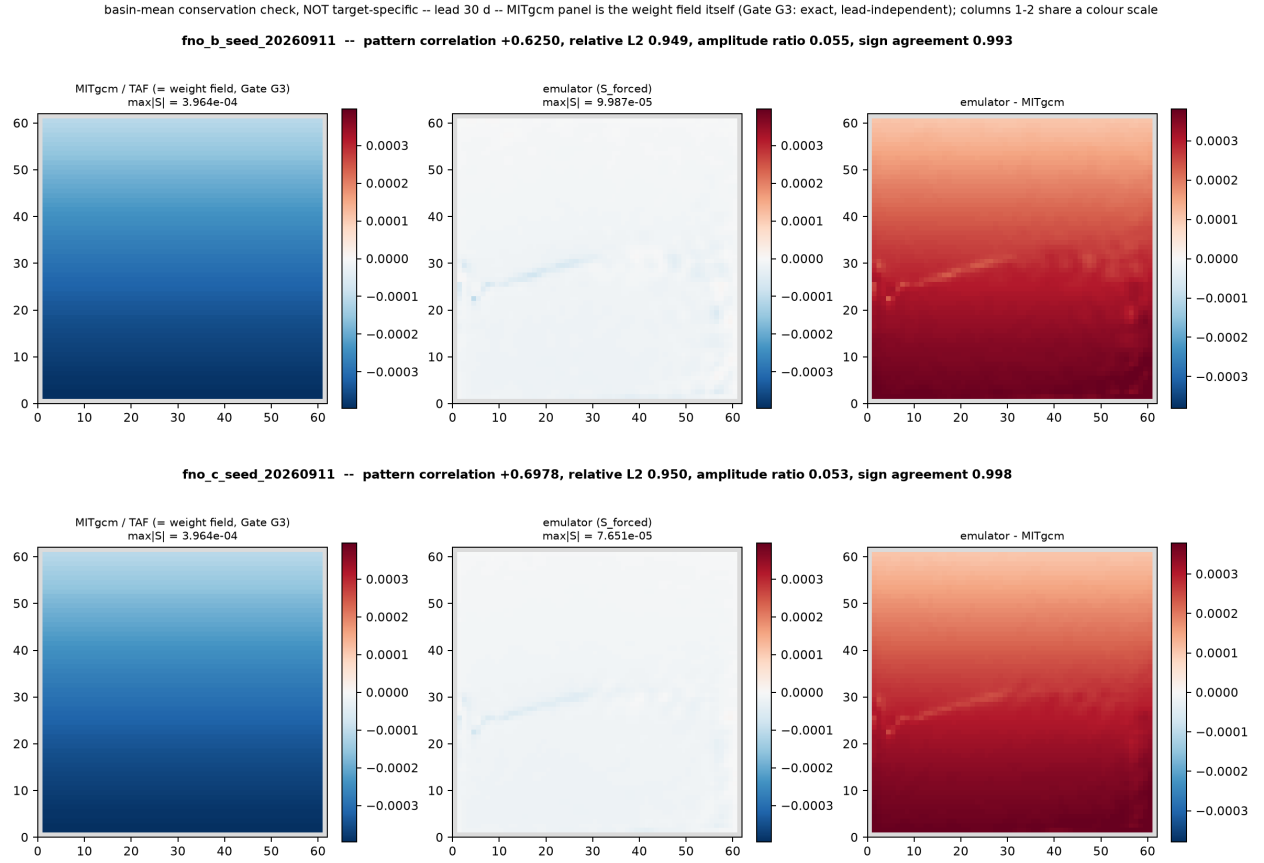


Figure 8: Basin-mean conservation, lead 30 d (three calls). The reference is unchanged; the emulator amplitude has fallen to $\approx 0.38^3 \approx 0.055$ of truth for both arms, and the pattern correlation has already dropped to 0.63–0.70.

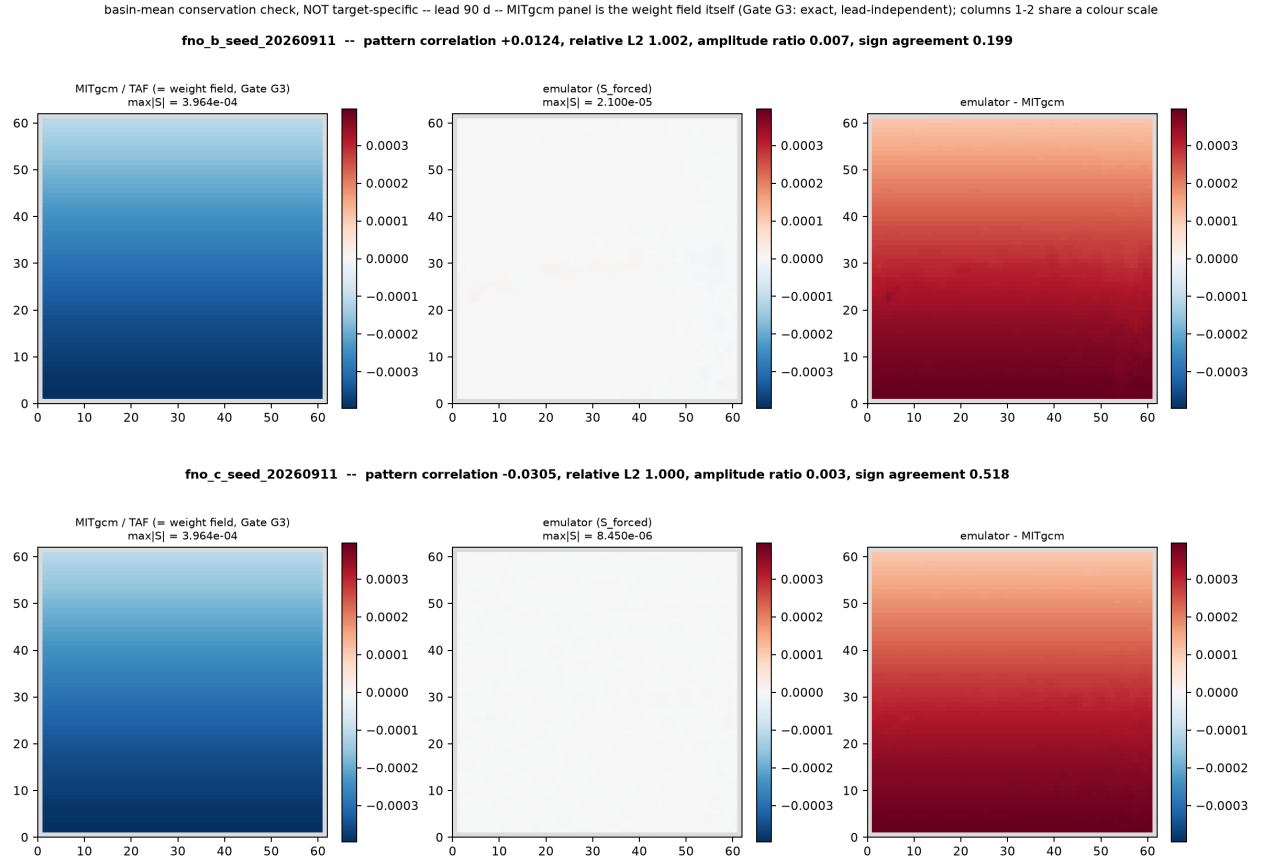


Figure 9: Basin-mean conservation, lead 90 d. The emulator panels are essentially blank on the shared scale: the conserved mode has been damped out of existence, and the difference panel is the reference field with its sign flipped.

4.2.b. Interior target

The interior target sits at $(i, j) = (31, 17)$, mid-basin, roughly half a domain width from either wall.

Objective	Lead (d)	Control			Response		
		ρ	ε_2	α	ρ	ε_2	α
J_{raw}	10	+0.125	8.330	8.363	+0.125	8.209	8.243
J_{raw}	30	+0.078	1.163	0.696	+0.082	1.152	0.682
J_{raw}	90	−0.121	1.016	0.108	−0.010	1.001	0.047
J_{anom}	10	+0.120	20.285	20.381	+0.121	19.991	20.086
J_{anom}	30	+0.049	2.115	1.913	+0.056	2.074	1.874
J_{anom}	90	−0.152	1.136	0.409	−0.001	1.016	0.177

Table 4: Interior target (31, 17). Bold marks the lower relative L_2 of the two arms; the response arm is lower in all six cells.

Raw SSH (Figures 10–12). The reference maps show what fast barotropic adjustment looks like in an adjoint. At lead 10 the true sensitivity is already basin-wide: a broad positive field everywhere — the conserved-mean pedestal of Eq. (14), which J_{raw} does not remove — with a strong positive lobe fanning east and north-east from the target and a weak negative notch just upstream of it. Ten days is long enough for a sea-level perturbation anywhere in the basin to have influenced the target. The emulator’s map is nothing like this. It is a near-delta at the target cell itself ($\max |S| = 1.12 \times 10^{-1}$ against 9.6×10^{-4} for truth, a factor of 116 at the peak) surrounded by fine-grained, sign-alternating speckle concentrated along the jet axis at $j \approx 30$ and against the eastern wall. The emulator’s ten-day Jacobian is, in effect, almost the identity plus grid-scale noise: it has learned that sea level ten days from now looks like sea level today at the same place, which is an excellent *forecast* and a completely wrong *derivative*. The resulting amplitude ratio is 8.4.

By lead 30 the two amplitudes have crossed: the true sensitivity has continued to spread and grow while the emulator’s has decayed, giving $\alpha = 0.70$ and $\varepsilon_2 = 1.16$. By lead 90 the reference is a smooth, everywhere-positive field with a broad maximum draped over the target and the jet, while the emulator retains only a small dipole at the target and traces along the jet, at 11% (control) and 4.7% (response) of the true amplitude. The pattern correlation is now slightly *negative*. Note that the two arms differ visibly here: the control keeps a stronger localized dipole ($\max |S| = 6.4 \times 10^{-4}$) than the response (1.2×10^{-4}), and the response’s lower ε_2 at this lead comes from being closer to zero, not from being closer to truth.

SSH anomaly (Figures 13–15). Removing the basin mean removes the pedestal and leaves the dynamically interesting part: at lead 90 the true map is a clean two-lobed structure — a broad positive region enveloping the target and extending south-east, a negative region hugging the western boundary — which is the imprint of westward Rossby propagation carrying information from the east back toward the target. The emulator maps show no

trace of it: the control has a small target-centred dipole plus jet-axis speckle, the response is nearly featureless. Because the reference norm is smaller once the pedestal is gone, the lead-10 amplitude ratios are correspondingly larger (20.4 and 20.1). The response arm's improvement is consistent but small: relative L_2 falls by 1.4%, 1.9% and 10.6% at leads 10, 30 and 90.

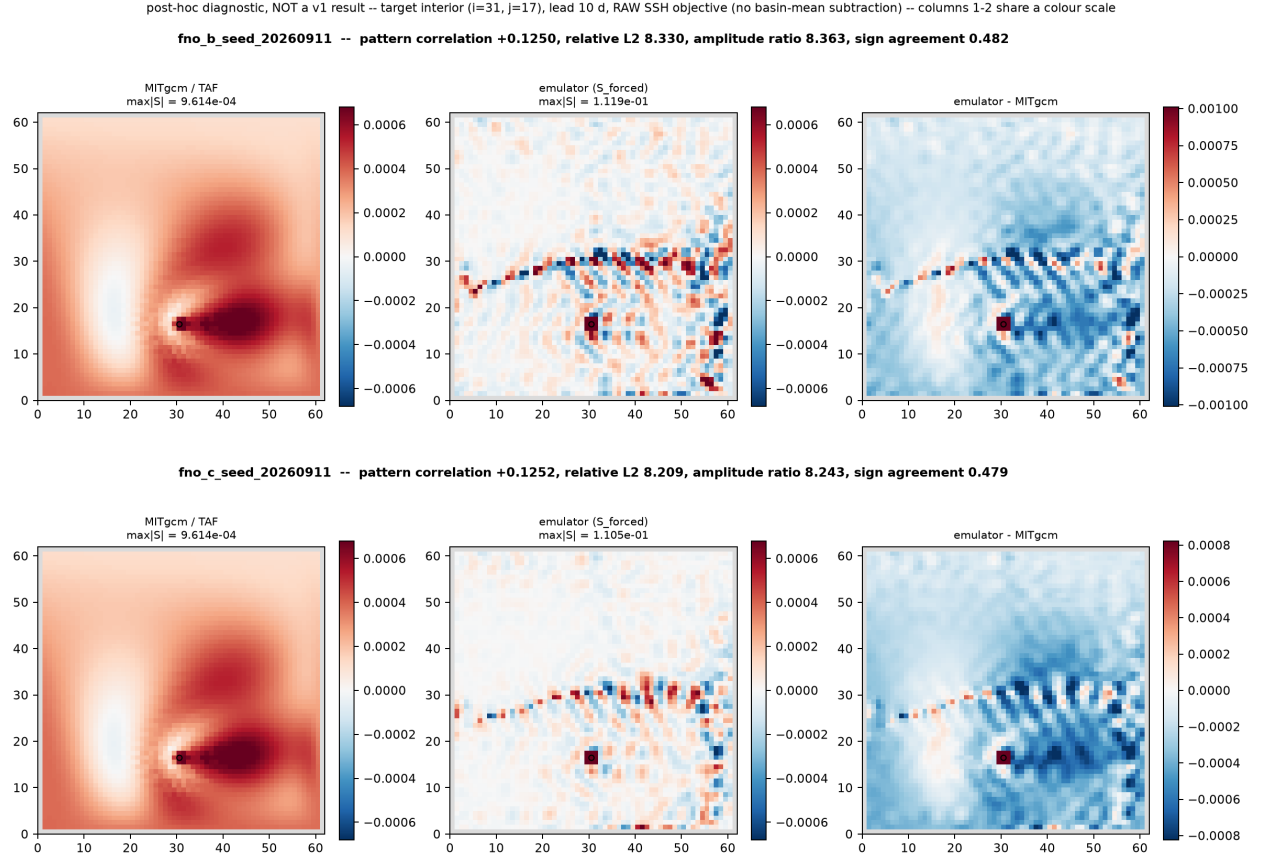


Figure 10: Interior target, J_{raw} , lead 10 d. MITGCM's sensitivity is already basin-wide; the emulator's is a near-delta at the target plus jet-axis speckle, $116\times$ too large at its peak.

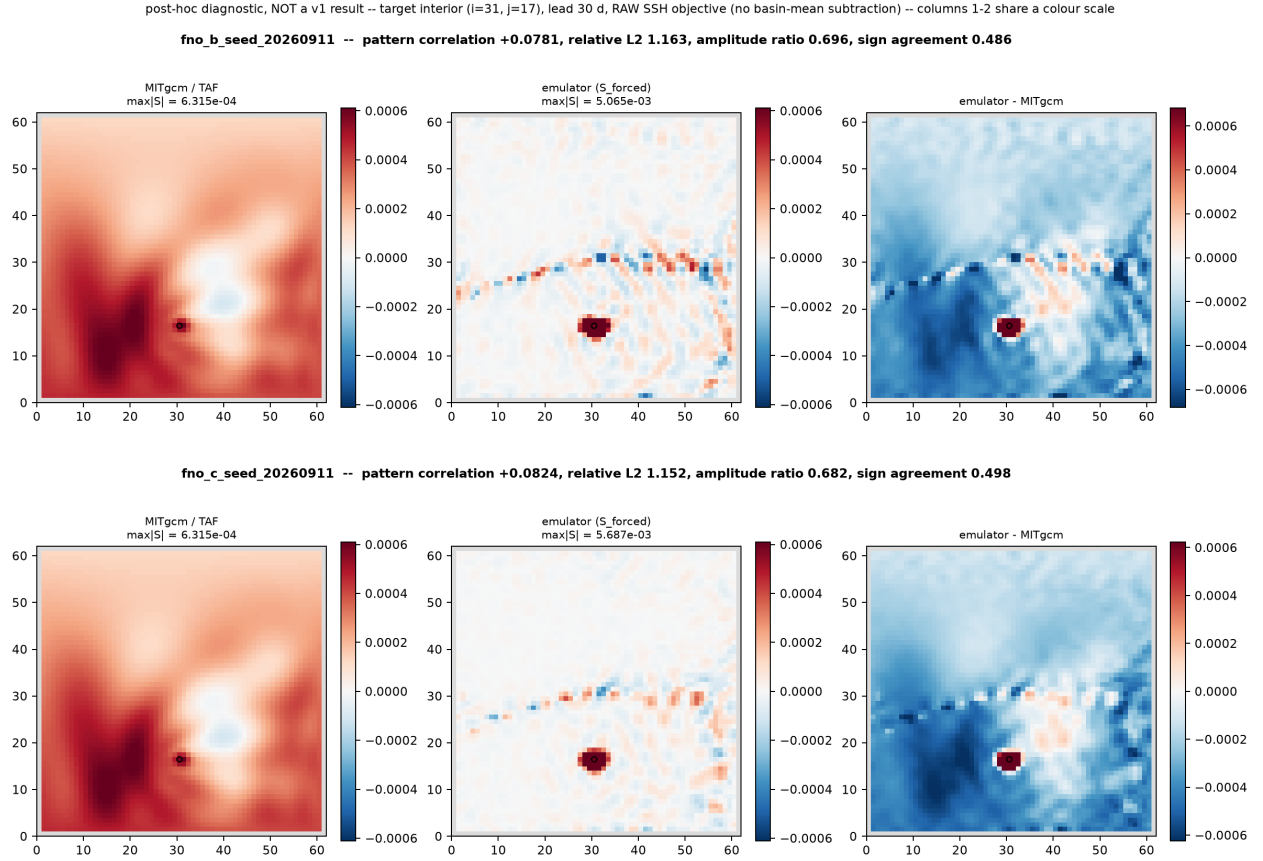


Figure 11: Interior target, J_{raw} , lead 30 d. The amplitudes have crossed: the true sensitivity keeps growing, the emulator's decays, giving $\alpha \approx 0.69$.

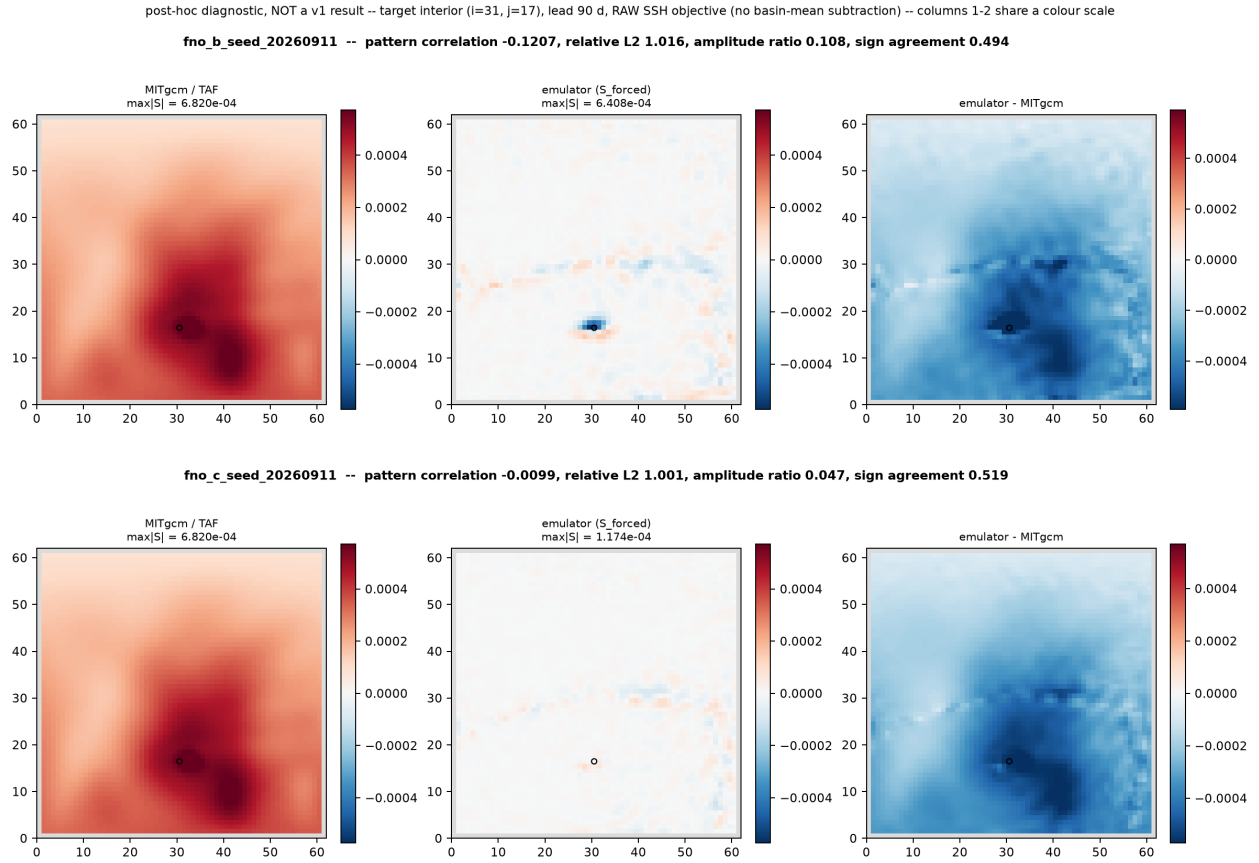


Figure 12: Interior target, J_{raw} , lead 90 d. The emulator retains 4–11% of the true amplitude and the correlation has turned slightly negative.

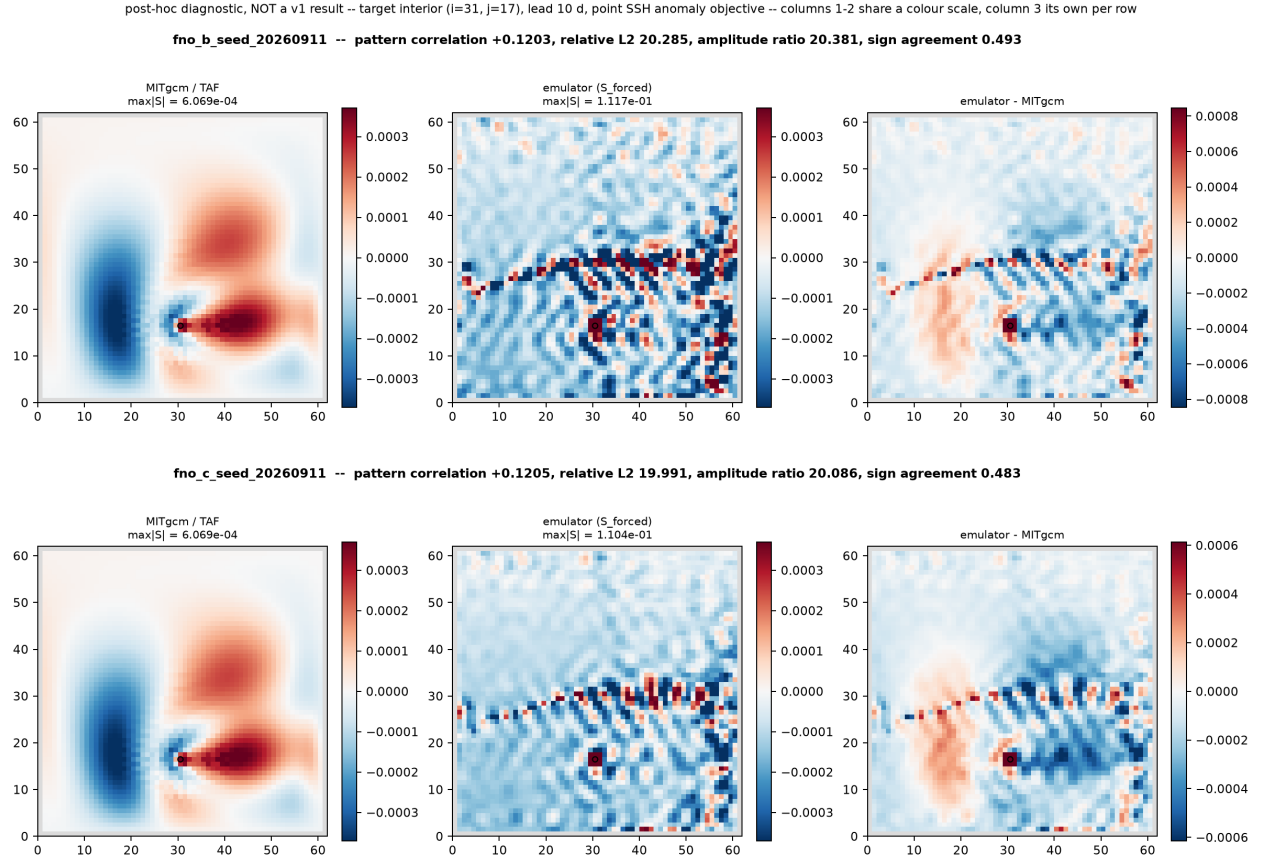
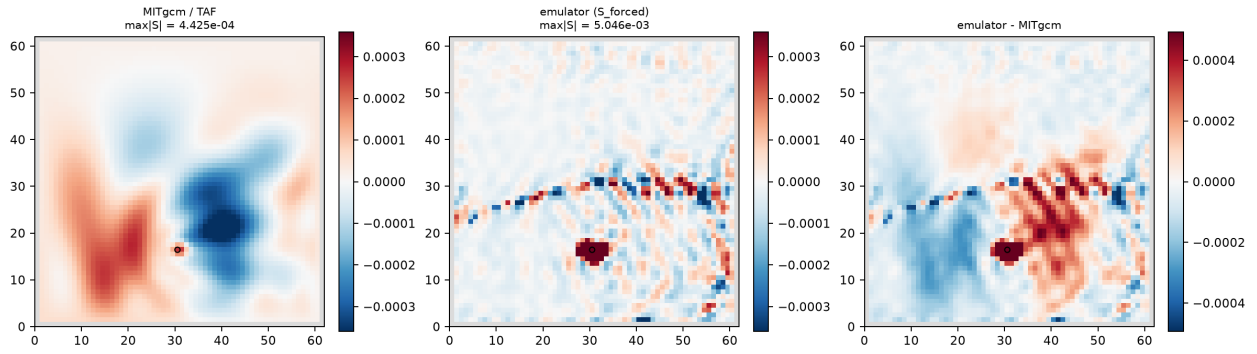


Figure 13: Interior target, J_{anom} , lead 10 d. With the conserved pedestal removed the reference norm is smaller, so the same emulator error registers as $\alpha \approx 20$.

post-hoc diagnostic, NOT a v1 result -- target Interior ($l=31, j=17$), lead 30 d, point SSH anomaly objective -- columns 1-2 share a colour scale, column 3 its own per row

fno_b_seed_20260911 -- pattern correlation +0.0487, relative L2 2.115, amplitude ratio 1.913, sign agreement 0.448



fno_c_seed_20260911 -- pattern correlation +0.0559, relative L2 2.074, amplitude ratio 1.874, sign agreement 0.419

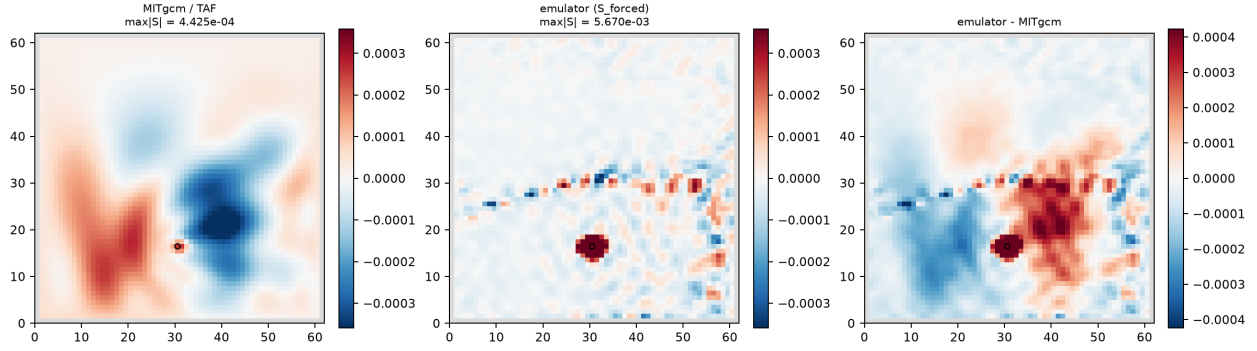


Figure 14: Interior target, J_{anom} , lead 30 d.

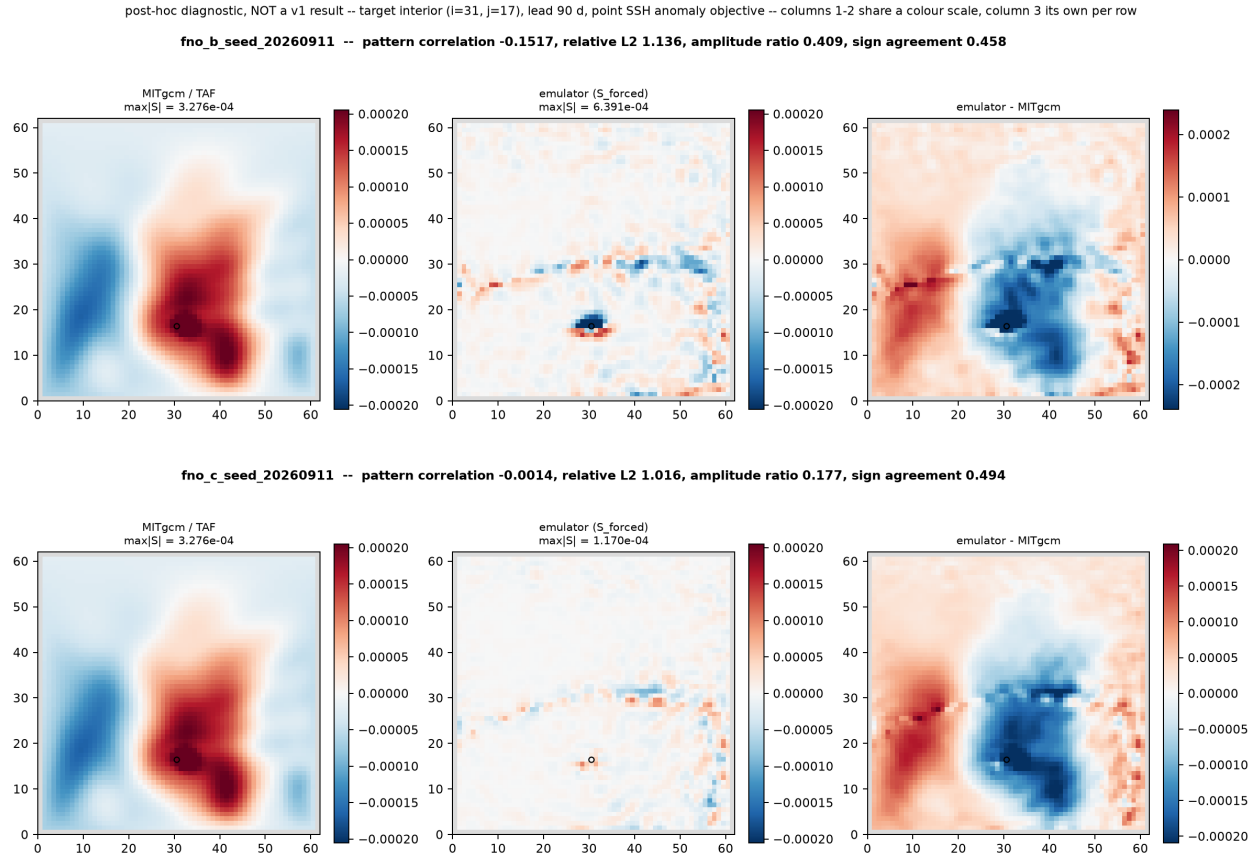


Figure 15: Interior target, J_{anom} , lead 90 d. The true map is a clean two-lobed Rossby-propagation pattern; neither emulator reproduces any part of it.

4.2.c. Eastern target

The eastern target sits at (61, 17), in the last wet column against the eastern wall.

Objective	Lead (d)	Control			Response		
		ρ	ε_2	α	ρ	ε_2	α
J_{raw}	10	+0.109	10.213	10.243	+0.108	9.835	9.863
J_{raw}	30	+0.109	1.257	0.857	+0.106	1.232	0.810
J_{raw}	90	−0.016	1.004	0.087	+0.037	0.999	0.026
J_{anom}	10	+0.178	47.016	47.183	+0.177	45.265	45.431
J_{anom}	30	+0.154	6.633	6.713	+0.154	6.264	6.339
J_{anom}	90	−0.034	1.466	1.038	−0.014	1.051	0.311

Table 5: Eastern target (61, 17). The response arm again has the lower relative L_2 in all six cells; the lead-90 anomaly cell is the largest single reduction in this report (28%).

Discussion. The eastern target produces the largest amplitude errors in the study — $47\times$ at lead 10 under J_{anom} — and also the highest pattern correlations, +0.18. Both facts have the same cause. The true lead-10 anomaly sensitivity (Figure 19) is a smooth, large-scale dipole: strongly positive over the entire south-western quadrant, negative through a broad lobe centred near (42, 20), with a thin band of alternating sign right against the eastern wall where the target sits. Its overall norm is small because the field is smooth and the eastern wall is the quietest part of the basin, so the emulator’s grid-scale error dominates the ratio. At the same time, the emulator’s map is broadly negative over most of the basin, which happens to agree in sign with a large fraction of the reference — hence $\sigma = 0.61$, the highest sign agreement anywhere in this report, and $\rho = +0.18$. That is coincidental large-scale bias agreement, not structural skill: the emulator’s map is dominated by a boundary-hugging spike at the target column ($\max |S| = 1.27 \times 10^{-1}$ against 4.5×10^{-4}) and by speckle along the jet, neither of which is present in truth.

The lead-90 anomaly cell is where the response term does the most good anywhere in this report: ε_2 falls from 1.466 to 1.051 and α from 1.038 to 0.311, a 28% reduction. But reading the two numbers together shows what happened. The control arm’s amplitude was already almost exactly right ($\alpha = 1.04$) and its correlation was -0.03 ; it had the right amount of sensitivity in entirely the wrong places. The response arm reduced the error by having *less* sensitivity ($\alpha = 0.31$), not by relocating it. That is the generic mechanism behind the improvements in these tables, and it is why a falling ε_2 alongside a flat ρ should not be read as progress toward a faithful adjoint.

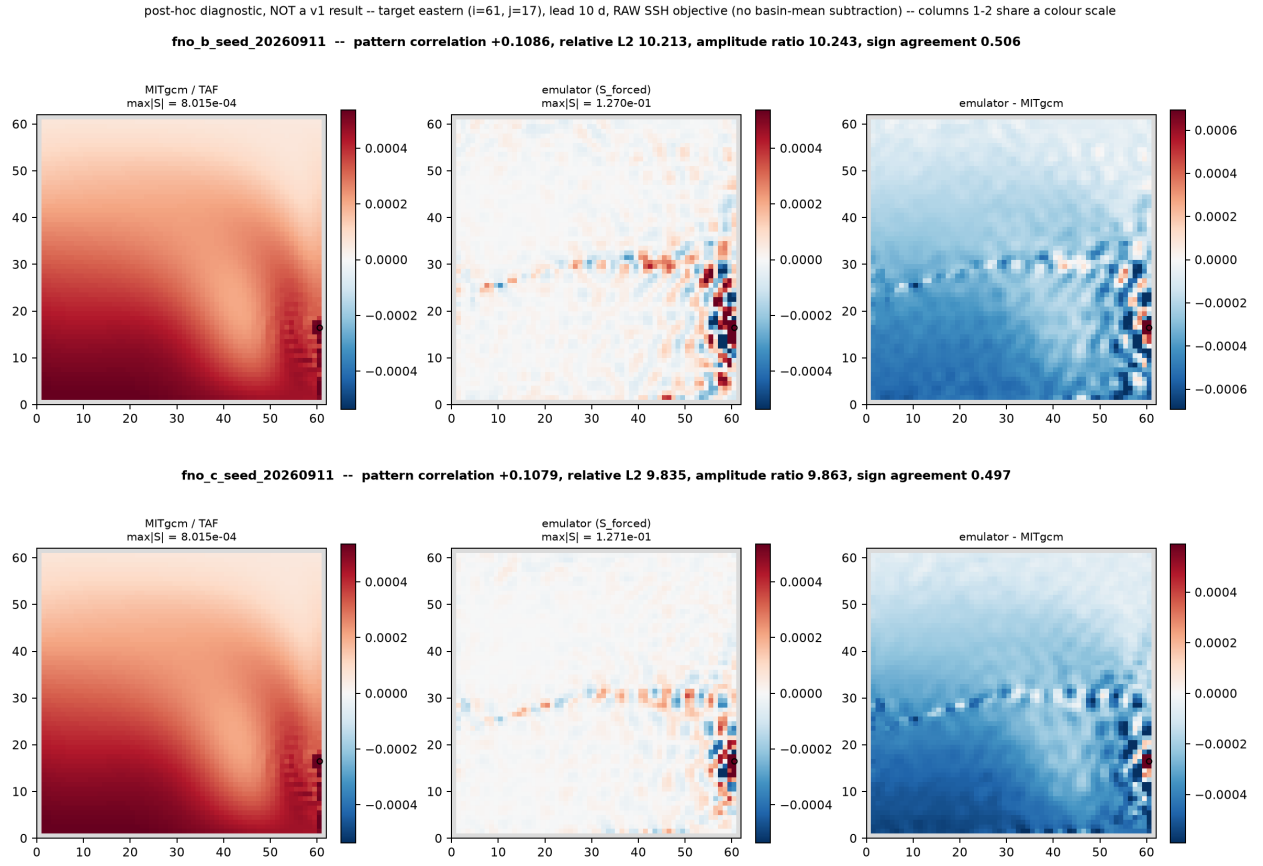


Figure 16: Eastern target, J_{raw} , lead 10 d.

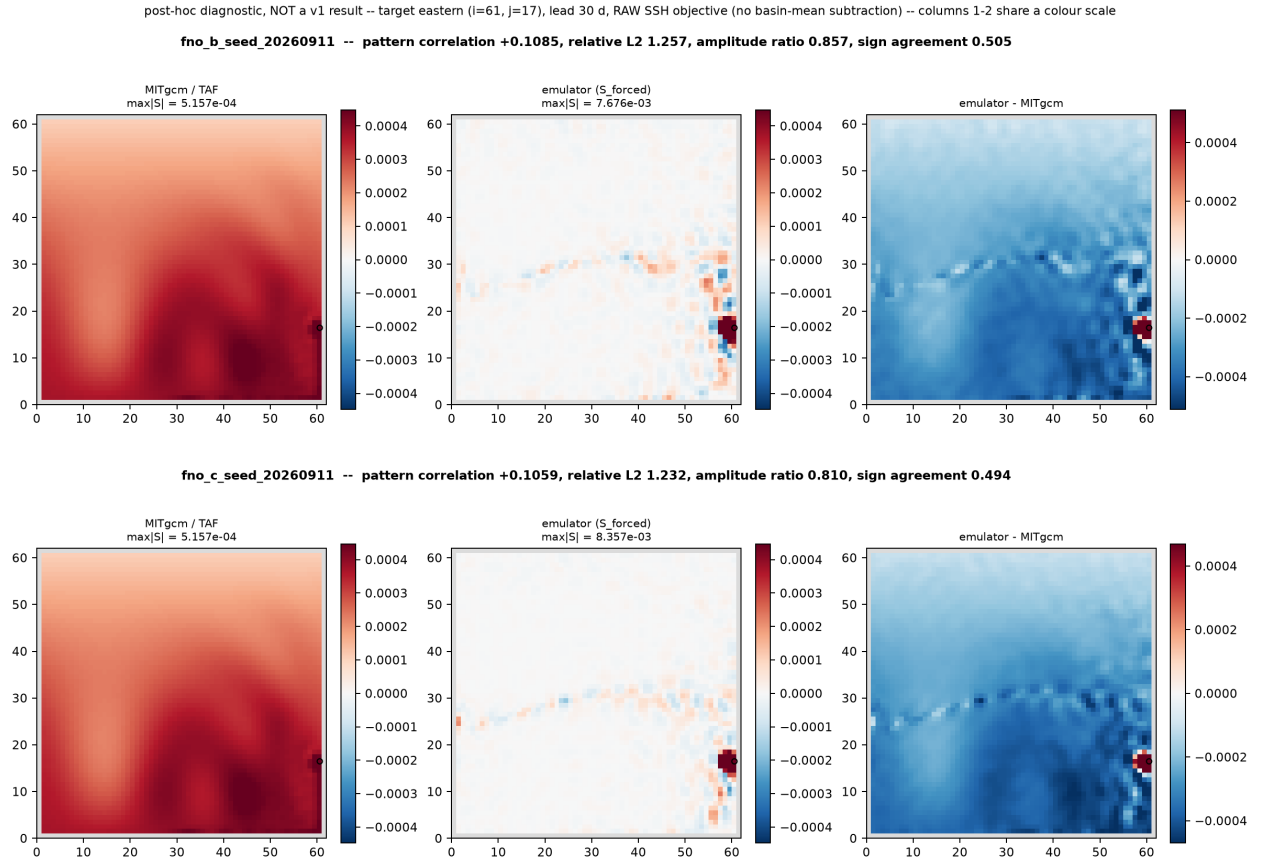


Figure 17: Eastern target, J_{raw} , lead 30 d.

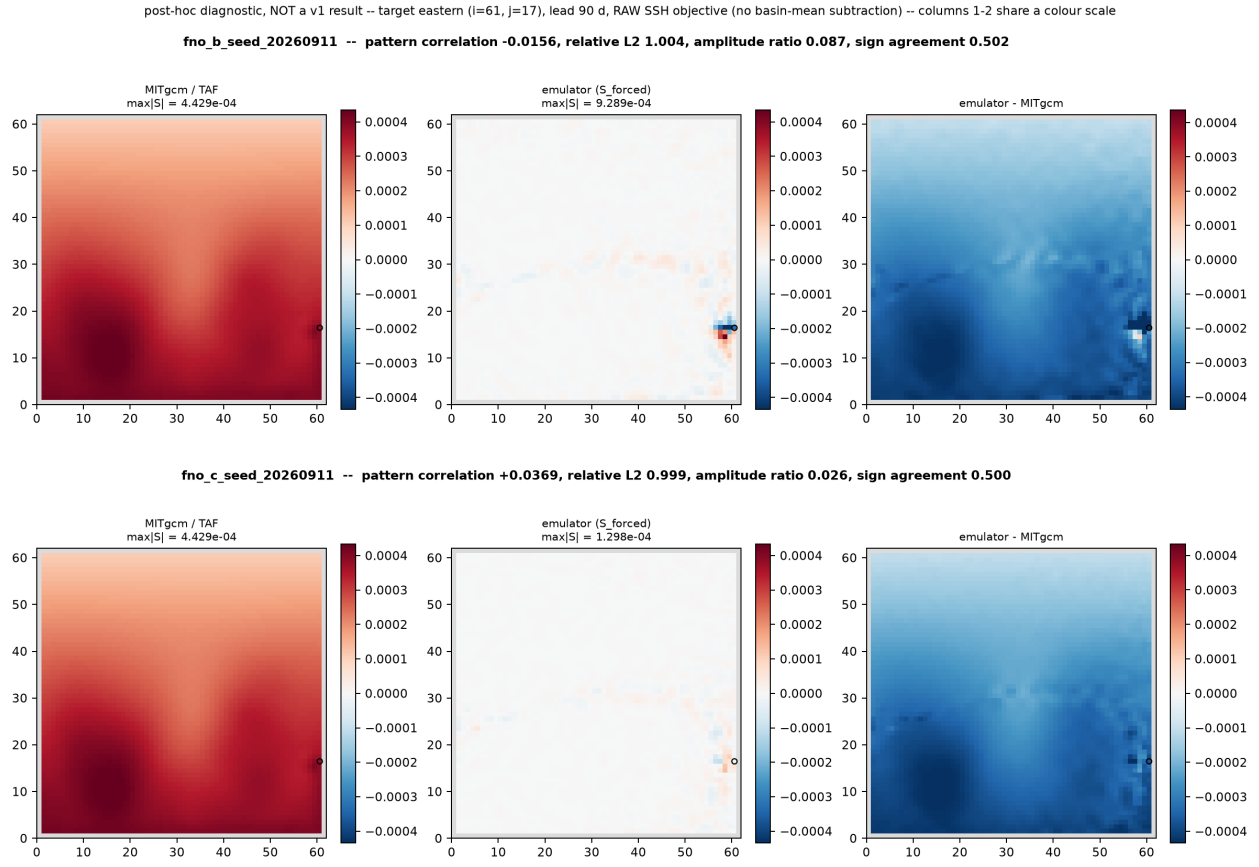


Figure 18: Eastern target, J_{raw} , lead 90 d. Both emulators have damped the signal to 3–9% of truth, and $\varepsilon_2 \rightarrow 1$ accordingly.

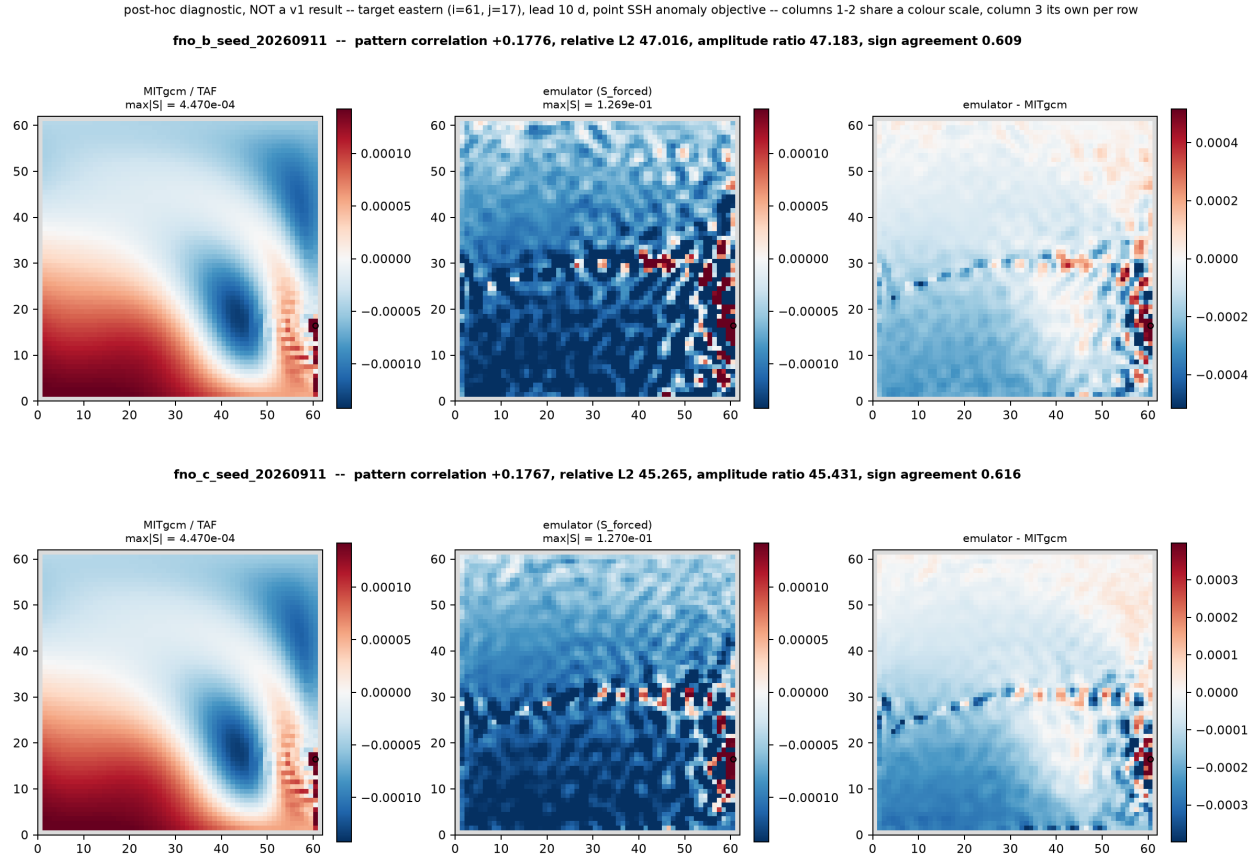
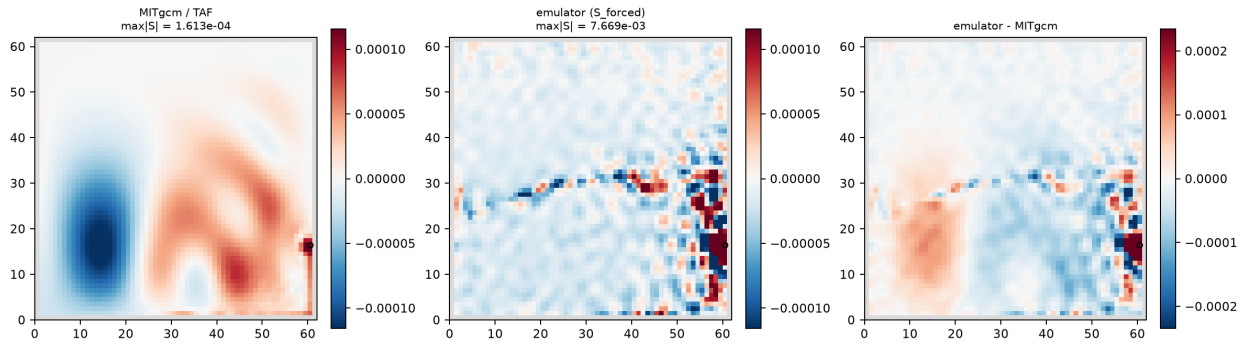


Figure 19: Eastern target, J_{anom} , lead 10 d — the largest amplitude error in this report ($\alpha \approx 47$). The reference is a smooth large-scale dipole with a small norm; the emulator is dominated by a boundary-hugging spike and jet-axis speckle.

post-hoc diagnostic, NOT a v1 result -- target eastern ($l=61, j=17$), lead 30 d, point SSH anomaly objective -- columns 1-2 share a colour scale, column 3 its own per row

fno_b_seed_20260911 -- pattern correlation +0.1542, relative L2 6.633, amplitude ratio 6.713, sign agreement 0.501



fno_c_seed_20260911 -- pattern correlation +0.1540, relative L2 6.264, amplitude ratio 6.339, sign agreement 0.481

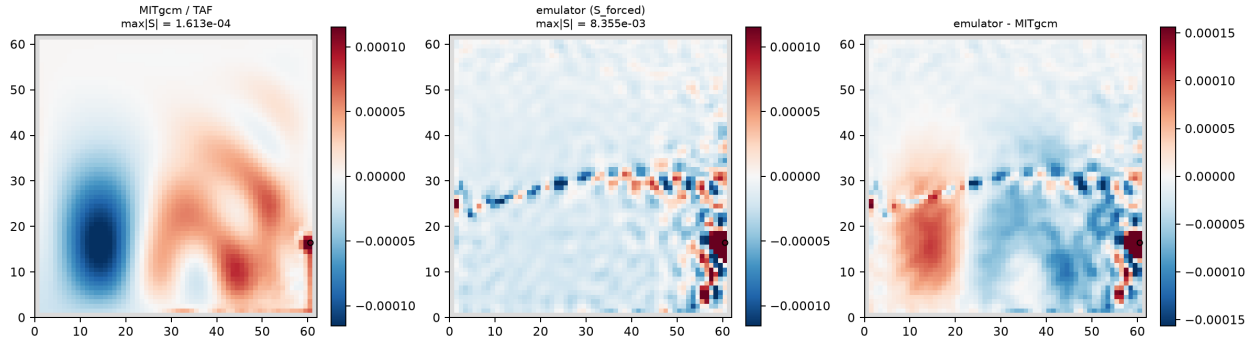


Figure 20: Eastern target, J_{anom} , lead 30 d.

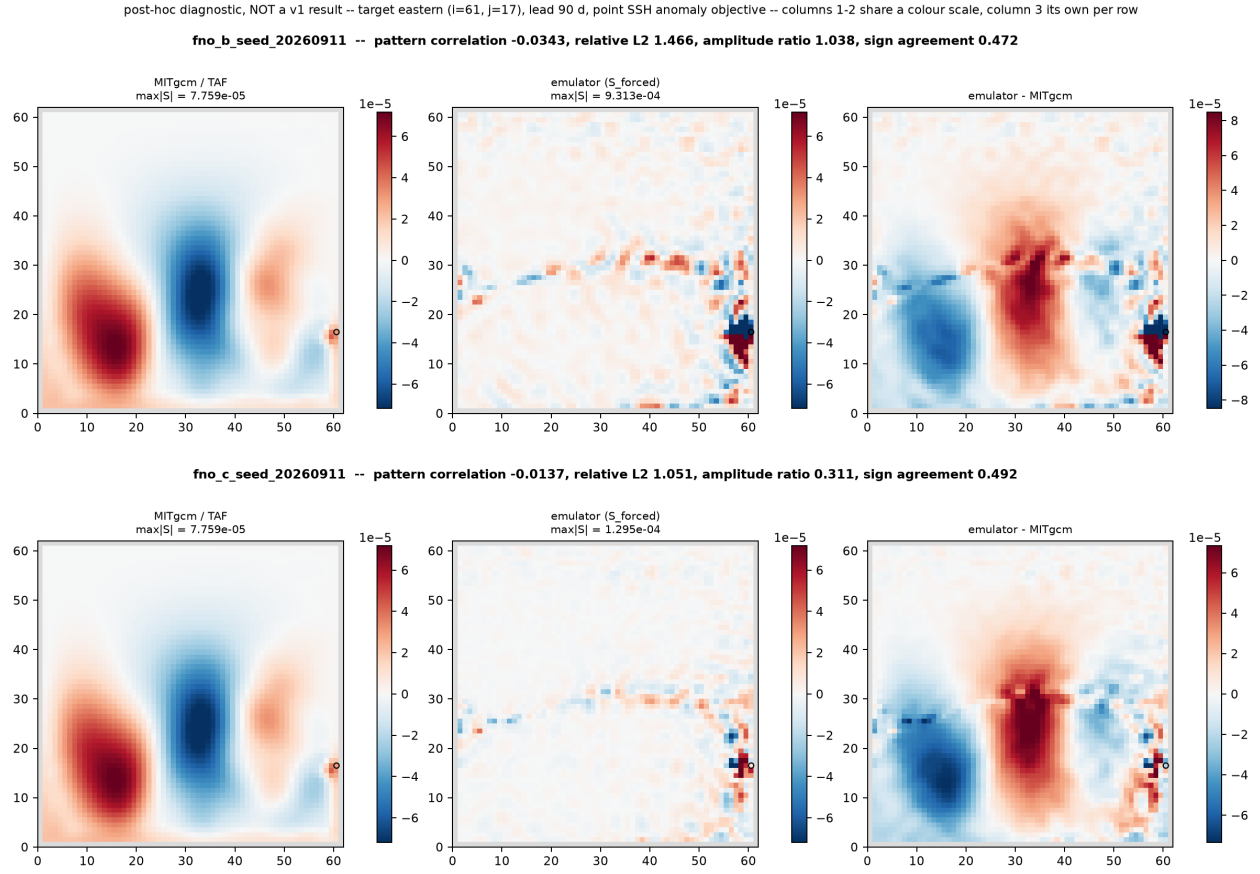


Figure 21: Eastern target, J_{anom} , lead 90 d. The control's amplitude is almost exactly right ($\alpha = 1.04$) with correlation -0.03 : the right amount of sensitivity in the wrong places. The response arm lowers the error by having less sensitivity, not by relocating it.

4.2.d. Western target

The western target is $p^* = (2, 17)$, the second wet column, inside the western boundary current. This is the study’s pre-declared confirmatory target, chosen because boundary-current sensitivity is what an ocean adjoint is normally used for and what a spectral emulator should find hardest.

Objective	Lead (d)	Control			Response		
		ρ	ε_2	α	ρ	ε_2	α
J_{raw}	10	+0.064	9.936	9.912	+0.071	8.814	8.794
J_{raw}	30	+0.015	1.563	1.224	+0.029	1.280	0.827
J_{raw}	90	−0.022	1.179	0.606	−0.023	1.115	0.455
J_{anom}	10	+0.051	40.854	40.892	+0.055	36.222	36.263
J_{anom}	30	+0.009	9.579	9.536	+0.027	6.500	6.450
J_{anom}	90	−0.030	7.063	6.962	−0.024	5.371	5.254

Table 6: Western target (2, 17), the confirmatory target. The response arm’s reductions in relative L_2 are the largest of the three targets under J_{anom} : 11%, 32% and 24% at leads 10, 30 and 90.

Discussion. The western target is where the failure is most visually stark. Under J_{anom} at lead 10 (Figure 25) the reference is smooth and organized: a positive tongue extending east from the boundary near $j \approx 20$, a deep negative pool centred near (46, 10), positive lobes on the northern rim and in the north-east, and a narrow band of fine structure in the first few columns at the target itself — the boundary current’s own signature. The emulator’s map, on the same colour scale, is saturated grid-scale noise across the entire basin, with alternating cells at $\pm$ the full scale. Its peak value is 1.47×10^{-1} against 3.2×10^{-4} for truth: a factor of 460 at the peak, 41 in L_2 . At lead 90 (Figure 27) the reference has organized itself into a clean pair of meridionally elongated lobes — exactly the westward Rossby signal an ocean adjoint should show — while the emulator has become a diagonal-striped checkerboard, still $7\times$ too large. This is the single clearest statement in the study that the emulator’s Jacobian is dominated by high-wavenumber content that the true Jacobian does not contain.

Under J_{raw} (Figures 22–24) the reference at lead 10 is almost uniformly positive: because the basin mean is exactly conserved, any perturbation anywhere raises the raw target value by roughly its share of that conserved total, on top of the local boundary structure. This is precisely the trivial signal that J_{anom} ’s mean subtraction was designed to remove, and comparing the two objectives at the same target and lead shows how much of the raw signal it accounts for: the amplitude ratio drops from 41 to 9.9 when the pedestal is left in, because the pedestal inflates the reference norm by roughly a factor of four while leaving the emulator’s grid-scale error unchanged.

The response arm’s gains are largest here — 32% at lead 30, 24% at lead 90, under the confirmatory objective — and it is the only target where the amplitude ratio crosses from above 1 to below 1 under J_{raw} (1.22 to 0.83 at lead 30). Yet the pattern correlation moves from +0.009 to +0.027 at lead 30 and stays negative at lead 90. The response term

is demonstrably doing something real and in a consistent direction; what it is doing is calibrating magnitude, not structure.

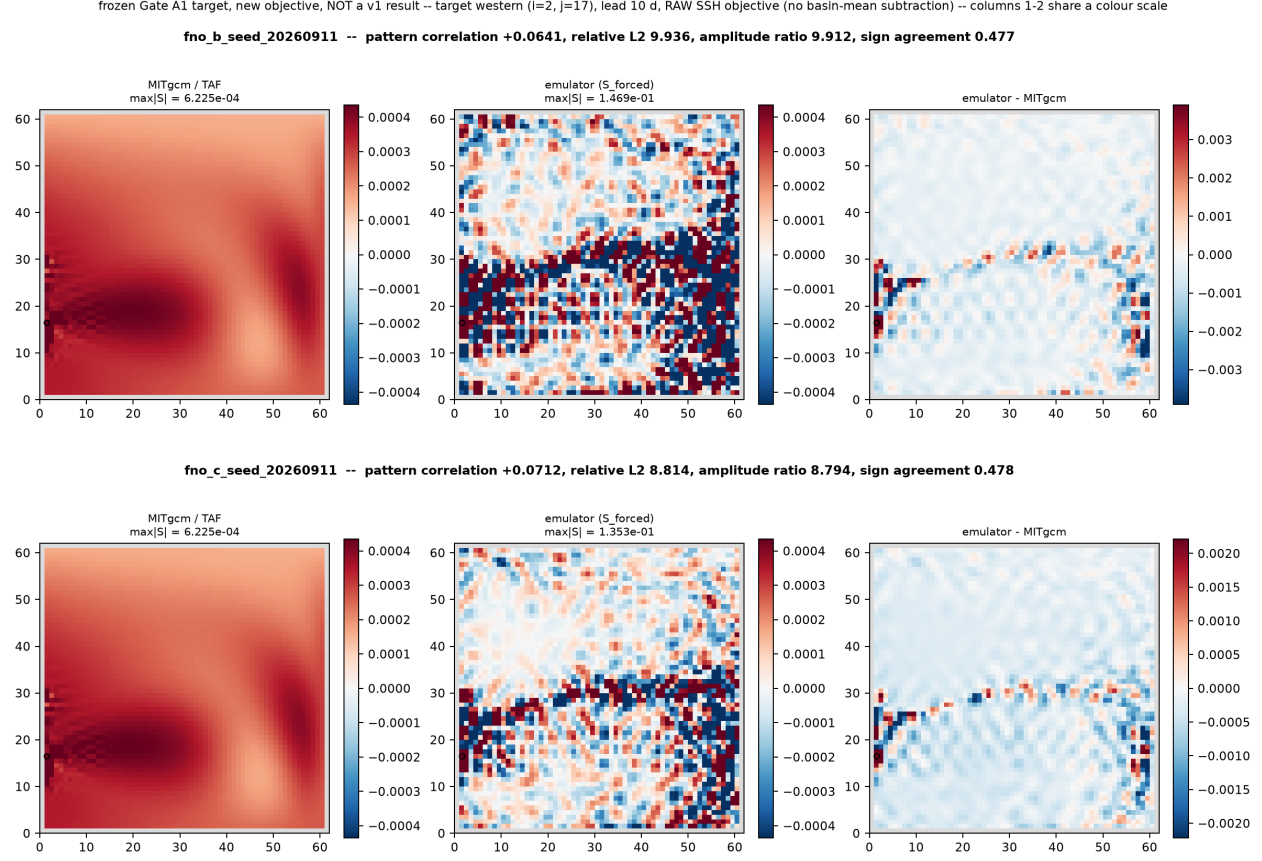


Figure 22: Western target, J_{raw} , lead 10 d. The reference is almost uniformly positive — the exactly conserved basin-mean pedestal plus local boundary structure — which is what the anomaly objective's mean subtraction removes.

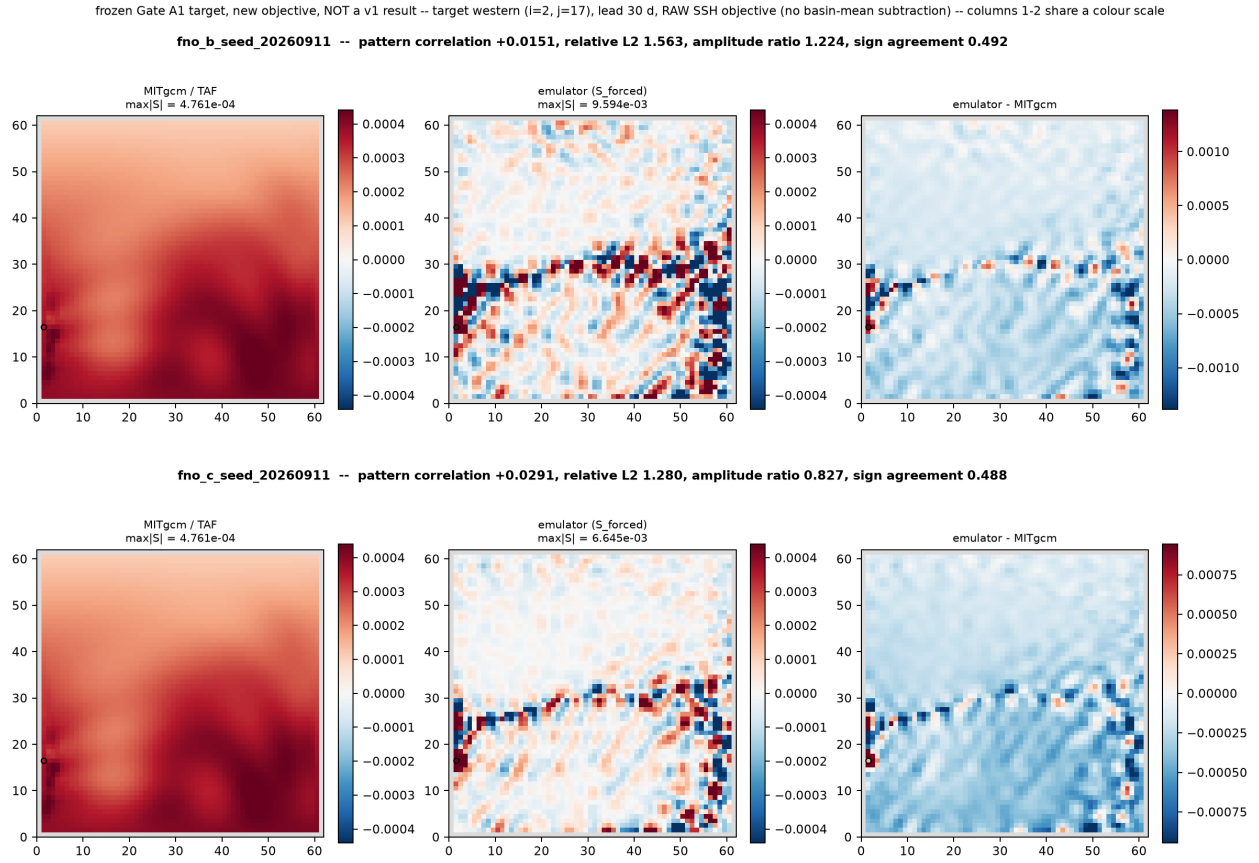
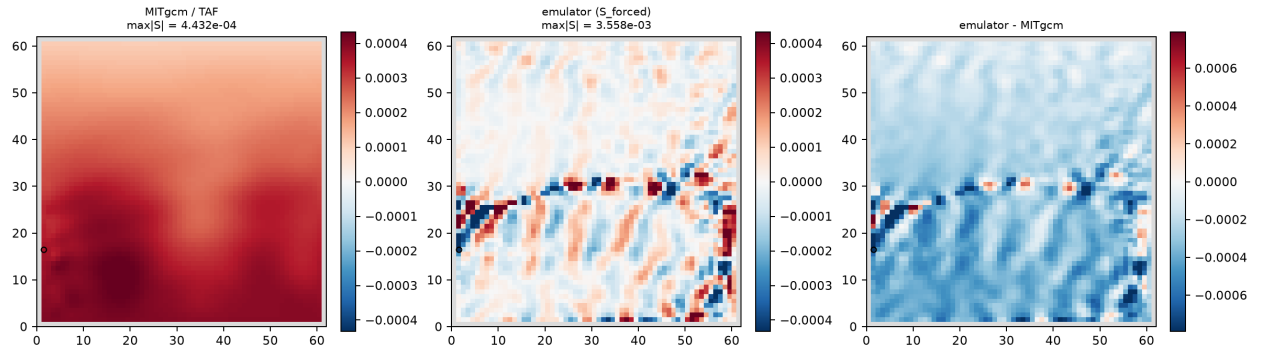


Figure 23: Western target, J_{raw} , lead 30 d. The only cell in this report where the response arm's amplitude ratio crosses from above to below 1 ($1.22 \rightarrow 0.83$).

frozen Gate A1 target, new objective, NOT a v1 result -- target western ($l=2, j=17$), lead 90 d, RAW SSH objective (no basin-mean subtraction) -- columns 1-2 share a colour scale

fno_b_seed_20260911 -- pattern correlation -0.0224, relative L2 1.179, amplitude ratio 0.606, sign agreement 0.489



fno_c_seed_20260911 -- pattern correlation -0.0225, relative L2 1.115, amplitude ratio 0.455, sign agreement 0.497

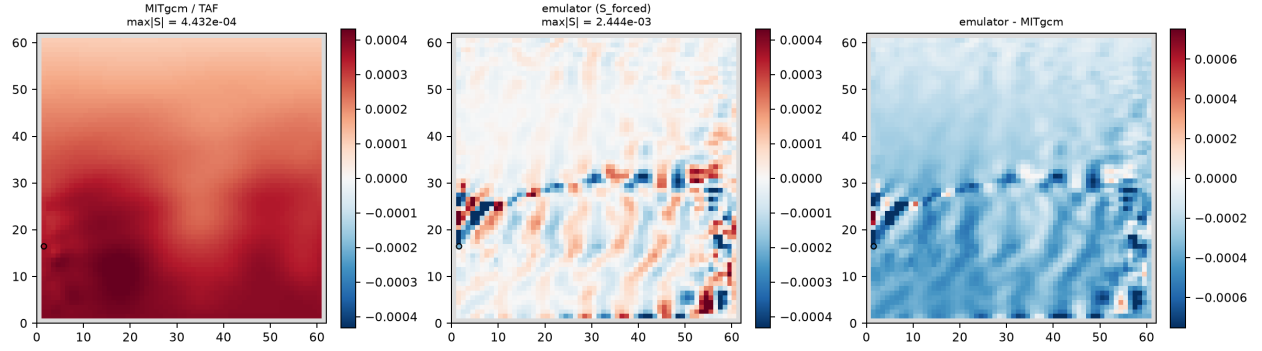


Figure 24: Western target, J_{raw} , lead 90 d.

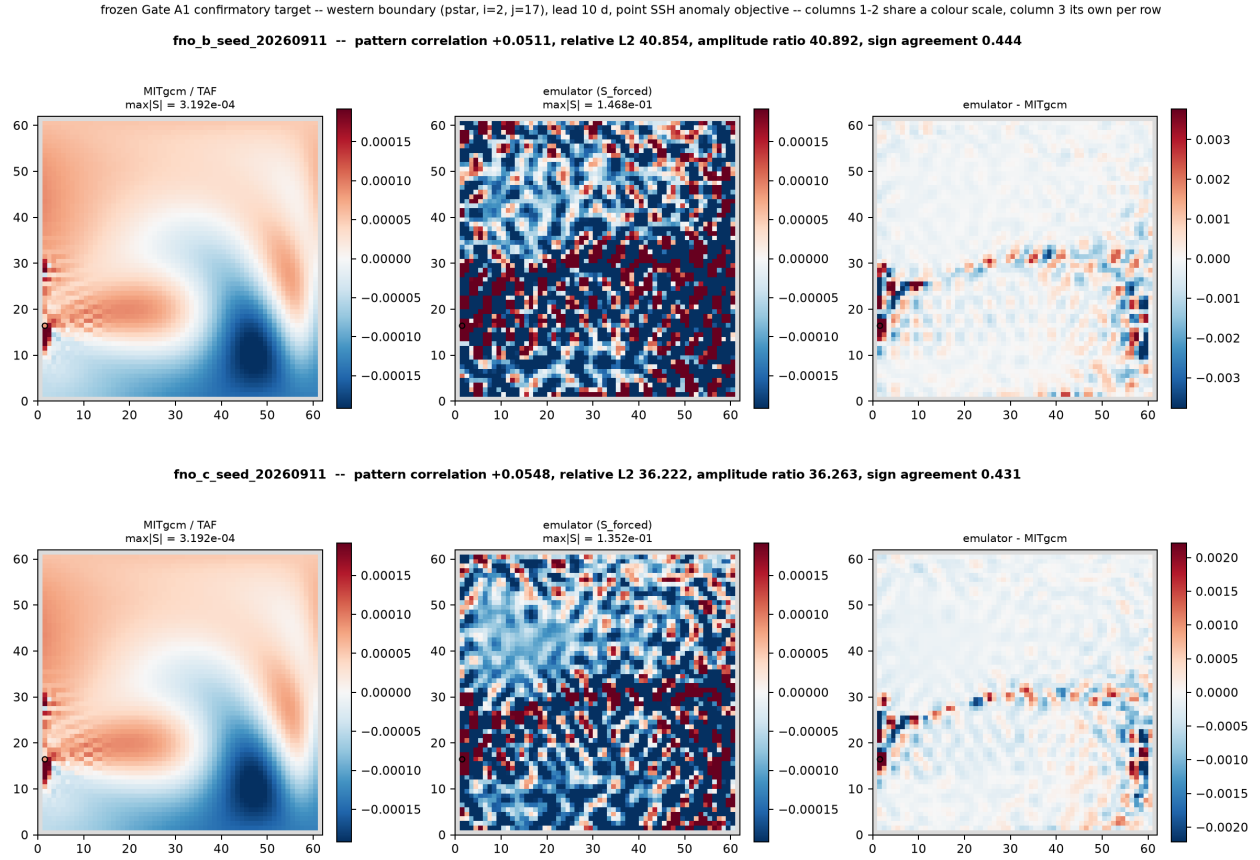
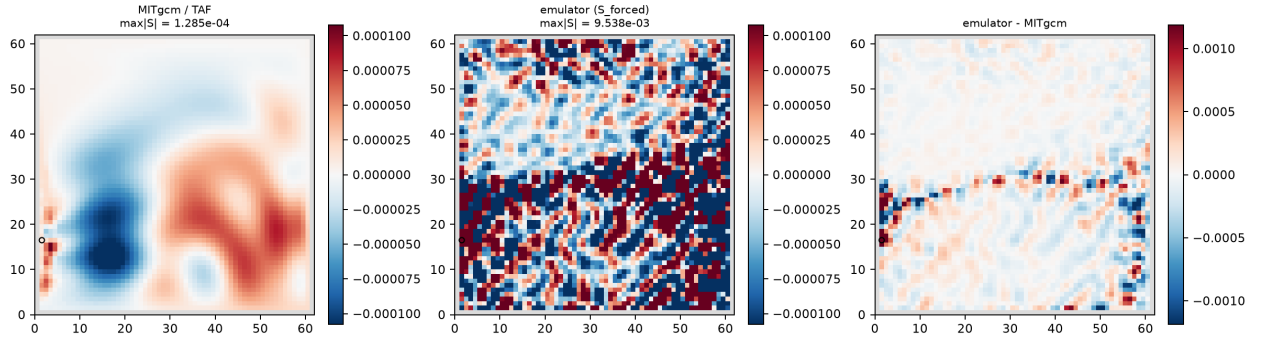


Figure 25: Western target, J_{anom} , lead 10 d — the study's confirmatory cell. On a shared colour scale the reference is smooth and organized while both emulator maps are saturated grid-scale noise, $460\times$ too large at the peak.

frozen Gate A1 confirmatory target -- western boundary (pstar, l=2, j=17), lead 30 d, point SSH anomaly objective -- columns 1-2 share a colour scale, column 3 its own per row

fno_b_seed_20260911 -- pattern correlation +0.0094, relative L2 9.579, amplitude ratio 9.536, sign agreement 0.514



fno_c_seed_20260911 -- pattern correlation +0.0272, relative L2 6.500, amplitude ratio 6.450, sign agreement 0.534

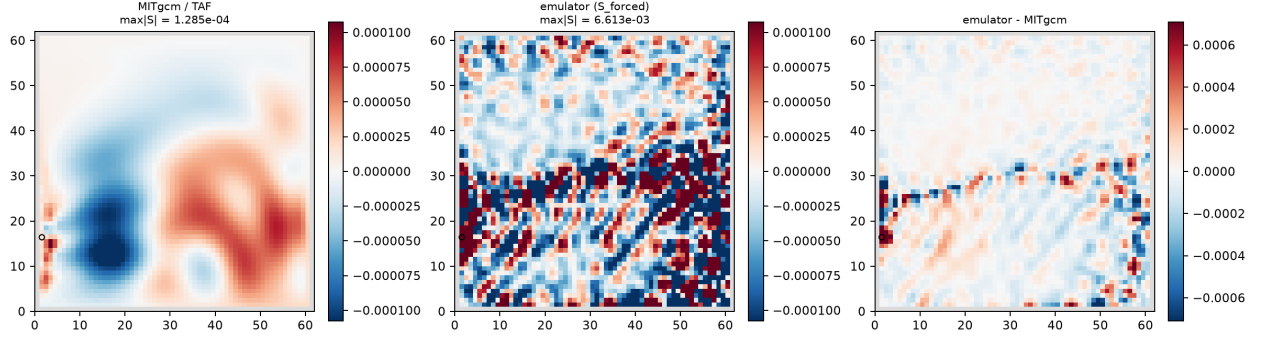


Figure 26: Western target, J_{anom} , lead 30 d. The largest single relative- L_2 reduction from the response term ($9.58 \rightarrow 6.50$, 32%), with pattern correlation still at +0.03.

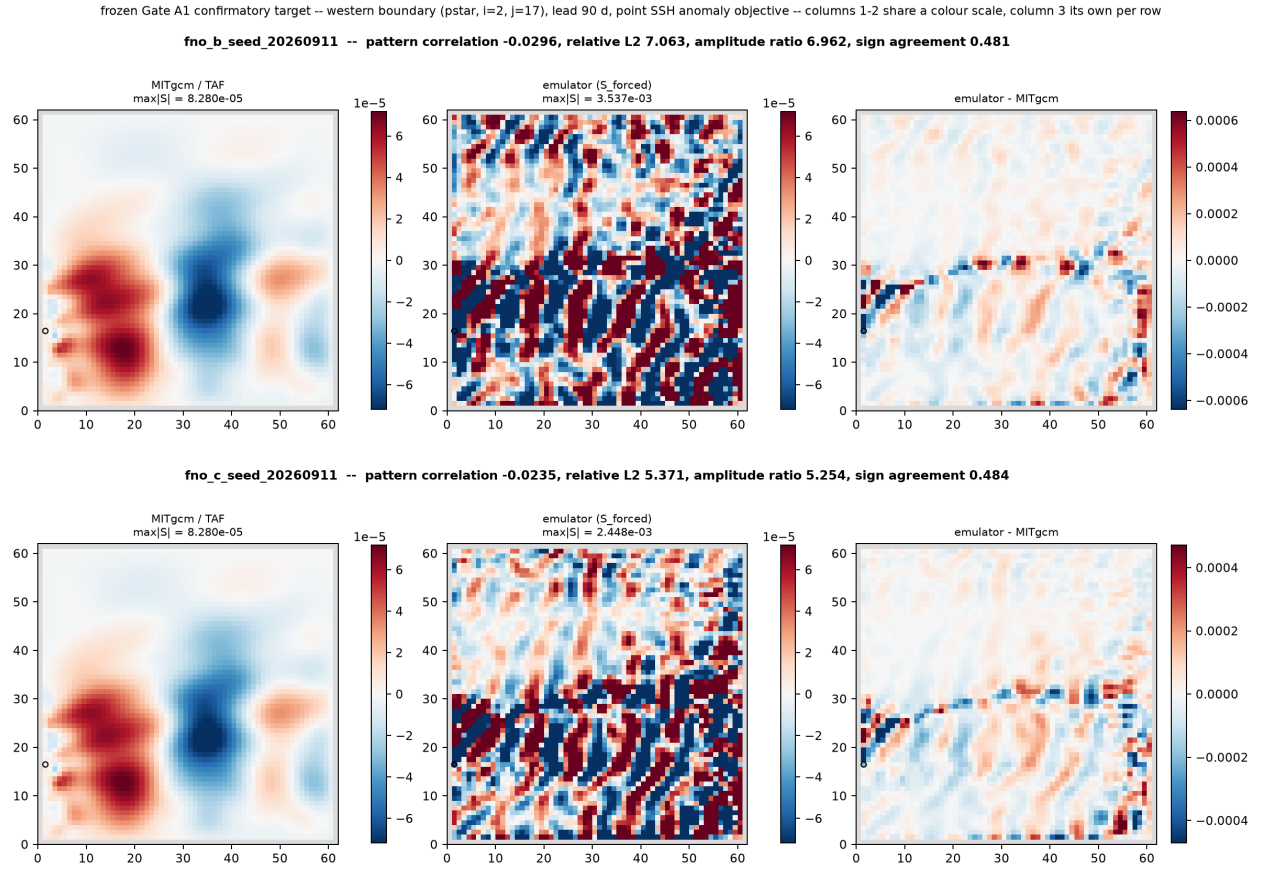


Figure 27: Western target, J_{anom} , lead 90 d. MITGCM's adjoint has organized into meridionally elongated Rossby lobes; both emulators show a diagonal-striped checkerboard, still $5\text{--}7\times$ too large.

5. Discussion

Three findings follow from Section 4, and they point in different directions.

The response term works, in the sense it was designed to work. Relative L_2 falls in all 18 (target, objective, lead) cells of Tables 4–6, by up to 32%. That consistency is not noise: the two arms share seed, data, split, schedule and every loss term but one, so the sign of the difference is attributable to L_{response} . The study’s pre-registered blind endpoint, computed on all three seeds before this report, agrees — relative L_2 fell in 24 of 24 confirmatory cells, with a median ratio of 0.745, and for seed 20260911 specifically the aggregate improvement was Δ_B ratio 0.743, comfortably inside the pre-declared ≤ 0.8 threshold. The gate nonetheless returned *negative*, because it is judged on the pre-declared primary seed 20260724, whose Δ_B ratio was 0.837. Five of the six gate criteria passed. Reporting seed 20260911 here does not change that verdict and is not intended to; it changes only which pair of models the figures show.

But it corrects magnitude, not structure. Pattern correlation is the metric the response term was supposed to move, and it does not move: across all 18 cells it stays between -0.15 and $+0.18$ for the control arm and between -0.03 and $+0.18$ for the response arm, with no systematic difference. Every improvement in Tables 4–6 is an amplitude improvement, and in several cells — eastern J_{anom} at lead 90 is the clearest — the response arm improves ε_2 by moving α *away* from 1, i.e. by producing a smaller wrong answer. The duality argument of Eq. (4) explains why this is possible: matching forward responses along a sampled set of directions constrains the adjoint’s projection onto that span, and with 672 directions in a state space of $46 \times 62 \times 62 \approx 1.8 \times 10^5$ dimensions, the span is a vanishing fraction of the whole. What the loss can cheaply satisfy is the overall scale of the response; what it cannot reach is the geometry of the Jacobian in the $\sim 10^5$ unsampled directions, which is what pattern correlation measures.

The failure has a consistent physical shape. Across every target and objective, the emulator’s ten-day sensitivity is a near-delta at the target cell plus grid-scale speckle along the jet axis and against the walls, whereas MITGCM’s is already basin-wide at ten days. That is the signature of an operator whose derivative is close to the identity: an excellent forecast rule for a slowly evolving ocean, and a wrong one for a basin in which barotropic adjustment crosses the domain in days. That gap is not a resolution limit of the ten-day step itself — MITGCM’s own adjoint reaches the whole basin within one such interval, and a spectral operator does not need a signal to physically cross the grid the way a local stencil would, since each retained Fourier mode already couples every grid point in a single application. The more likely explanation is architectural. Of the operations that make up each block — a per-mode spectral convolution, a linear channel-mixing term applied identically at every grid point, a channel-wise multilayer perceptron, and pointwise layer normalization — only the spectral convolution can connect two different grid cells at all; the rest, together with the lifting and projection layers bracketing the three blocks, act independently at each point and so can only ever contribute to the map’s response at its own cell, never

to its spread. The bias-free local correction added at the end is local by construction, coupling each cell only to its immediate few neighbours. A derivative built mostly from these pointwise and local pieces, with the spectral path’s low-wavenumber content weighing in only weakly, is exactly a near-delta with a narrow high-wavenumber halo — which is what is observed. A related diagnostic on this project’s production checkpoint, which shares this architecture, is consistent with that picture without establishing it for the two arms studied here: halving the local correction’s magnitude at inference cut the long-horizon spurious anomaly’s high-wavenumber power 24-fold, at a cost of 3.9-fold worse short-lead surface-pressure error — evidence that this one pathway carries a large share of both the short-lead skill and the long-horizon artefact. Nothing in the eight-term forecast objective constrains which pathway carries the derivative, since it never evaluates the derivative at all and rewards a near-persistent map cheaply at short lead, so training has no reason to shape the one pathway capable of the smooth, basin-wide coupling MITGCM’s adjoint shows; forward response data supervised at the same ten-day cadence inherits that same blindness. The mean-conservation result (Section 4.2.a) is the counterpart at the other extreme: a mode with true amplification exactly 1 is damped by a near-constant 0.38 per call, which is exactly what a stability regularizer designed to prevent autoregressive blow-up will do to a neutral mode. Both emulators are simultaneously far too sensitive at high wavenumber and far too damped at the largest scale.

Forward cost. The response term is not free. On this seed, at day 2,000 the response arm’s error exceeds climatology in all three primary fields while the control arm’s does not, its worst-case normalized amplitude rises from 6.55 to 9.89 (past the declared ceiling of 8), and its member spread at day 2,000 widens by more than an order of magnitude. Across the three seeds the sign of the paired forward difference flips, so this is not a systematic degradation attributable to L_{response} — seed-to-seed variance in long-rollout behaviour is larger than the treatment effect. That variance is itself the more important observation: a training recipe whose day-2,000 climatology ratio ranges from 0.22 to 5.79 across three seeds is not yet controlling the property it needs to control, and no adjoint-fidelity conclusion drawn from a single seed of such a recipe would be safe. It is the reason the study’s frozen contract requires three paired seeds and forbids selecting a best one.

6. Conclusion

A matched pair of FNO emulators, identical except for one added forward-perturbation-response loss term, was trained on the weakly turbulent S0 double gyre and compared both forward and against MITGCM/TAF adjoint sensitivities at three targets, two cost functionals and three leads. Forward, the two arms follow the same evolution to roughly day 700 and then separate, with the control arm remaining below climatology to day 2,000 on this seed and the response arm crossing it. In the adjoint, the response term reduces the sensitivity error consistently — relative L_2 falls in every cell measured, by up to 32% — but leaves pattern correlation flat near zero everywhere. The emulator’s ten-day Jacobian remains near-identity plus grid-scale noise where the true one is already basin-wide, and both arms

damp the exactly conserved basin-mean sea-level mode by a factor of ≈ 0.38 per ten-day call.

The conclusion is that matching forward response magnitudes is insufficient to recover adjoint structure. A response loss over a few hundred sampled directions constrains the Jacobian’s projection onto that span, and the scale of the response is the cheapest way to satisfy that constraint; the geometry of the operator in the remaining $\sim 10^5$ directions is untouched. A direct follow-up should first separate each trained model’s ten-day raw-SSH sensitivity into the contributions of the external 3×3 local branch and the Fourier backbone, evaluated at the same MITGCM state. This diagnostic would test whether removing the local contribution improves spatial agreement with the reference adjoint, rather than merely reducing sensitivity amplitude. The next controlled experiment would retrain both the control and response models without the 3×3 branch, keeping each arm’s data, losses, training schedule and budget matched to its original counterpart. Retraining would allow the backbone to compensate for the removed branch. Evaluation against independent MITGCM/TAF adjoints should assess spatial structure, sign, amplitude and mean-mode conservation, alongside forward accuracy and long-rollout stability. If the backbone alone retains the poor sensitivity structure, removing the local branch will be insufficient. This comparison should use multiple paired seeds under a new pre-registered cycle, so that any improvement can be distinguished from the substantial seed-to-seed variation documented here.

Data Availability

The code, configurations and selected figures for both models discussed in this report are available at <https://github.com/m-jalabert/adjoint-faithful-fno>. Both the control model (arm B, `model_c_adjoint_faithful_nominal_control_v1`) and the response model (arm C, `model_c_adjoint_faithful_response_v1`) are available on the `main` branch. The former `adjoint-faithful-response-training-v1` branch has been incorporated into `main` and removed. Commit `94137c39fa8aca62a7345f816f0a0b33f12e84cd` records the code and configurations used by this report and provides a fixed reference for reproduction. Raw MITGCM trajectories, restart pickups, processed datasets and trained checkpoints are stored separately on HPC scratch and are not included in Git. Readers must regenerate the trajectories and train the models, or obtain those artifacts separately. The repository’s `docs/ADJOINT_HANDOVER.md` describes the required environment, data regeneration, training and evaluation steps.

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Appendix A. Verifying the two sensitivity pipelines

This appendix describes how each of the two sensitivity maps compared in Section 4.2 is produced and independently checked, for a reader who did not write either pipeline and wants to know what stands behind a number in Tables 3–6 before trusting it.

Appendix A.1. S_{forced} versus S_{free} : which state the Jacobian is evaluated at

Both variants are built from the identical trained one-step Jacobian DF_θ and the identical reverse-mode machinery; they differ only in which states the interior links of the chain rule are evaluated at, not in how the derivative is computed.

Write x_0 for the shared true initial state at day 7200, $x_1^M, \dots, x_k^M$ for MITGCM’s true states at the following ten-day steps, and $x_0^F = x_0$, $x_{j+1}^F = F_\theta(x_j^F)$ for the emulator’s own autoregressive iterates. Because the cost is linear in the final state, $J(x_k) = w \cdot x_k[\eta]$, the seed $\nabla_{x_k} J$ does not depend on x_0 , and reverse accumulation through the k -fold composition $F_{\theta,k} = F_\theta \circ \dots \circ F_\theta$ gives a product of k one-step Jacobian transposes,

$$\frac{\partial J}{\partial x_0} = DF_\theta(x_0)^\top DF_\theta(x_1)^\top \dots DF_\theta(x_{k-1})^\top \nabla_{x_k} J. \quad (\text{A.1})$$

The only freedom left in Eq. (A.1) is which states $x_1, \dots, x_{k-1}$ the $k - 1$ interior factors sit at, and S_{forced} and S_{free} answer that differently.

$$S_{\text{free}} = DF_\theta(x_0^F)^\top DF_\theta(x_1^F)^\top \dots DF_\theta(x_{k-1}^F)^\top \nabla_{x_k} J, \quad x_{j+1}^F = F_\theta(x_j^F), \quad (\text{A.2})$$

evaluated along the emulator’s own forecast, is exactly $DF_{\theta,k}(x_0)^\top \nabla_{x_k} J$ — the true total derivative of the deployed k -step autoregressive map, because x_j^F really is what F_θ produces at step j . This is `free_chain`: one forward rollout, then a single `torch.autograd.grad` call through the whole composed graph.

$$S_{\text{forced}} = DF_\theta(x_0^M)^\top DF_\theta(x_1^M)^\top \dots DF_\theta(x_{k-1}^M)^\top \nabla_{x_k} J, \quad (\text{A.3})$$

where x_j^M is MITGCM’s own true state at step j , supplied externally at every step rather than generated by iterating F_θ . This is not the Jacobian of any single map that could be evaluated forward — one cannot autoregress “on truth” without external correction at

every step — it is k independently evaluated one-step Jacobians of F_θ , chained by matrix multiplication exactly the way MITGCM’s own adjoint chains k one-step Jacobians of the true dynamics along the true trajectory: TAF linearizes the nonlinear model about its own nonlinear forward trajectory at every timestep, by definition of an adjoint. This is `forced_chain`: k separate `torch.autograd.grad` calls, one per step, each seeded from the archived truth state rather than from the previous step’s emulator output.

F_θ is a nonlinear operator, so DF_θ genuinely depends on where it is evaluated: $DF_\theta(x_j^F) \neq DF_\theta(x_j^M)$ whenever $x_j^F \neq x_j^M$. Since the emulator’s own forecast drifts from MITGCM truth as j grows — exactly the forward RMSE of Table 2 — the interior factors in Eqs. (A.2)–(A.3) are evaluated at increasingly different points for $k \geq 2$, and

$$S_{\text{forced}} - S_{\text{free}} = \left[\prod_j DF_\theta(x_j^M)^\top - \prod_j DF_\theta(x_j^F)^\top \right] \nabla_{x_k} J \quad (\text{A.4})$$

is a direct measure of that trajectory divergence propagated through $k - 1$ Jacobian factors, not a difference in how either side’s derivative is computed. At $k = 1$ (lead 10) there is a single factor and $x_0^F = x_0^M = x_0$ trivially, so the two are identical bit for bit — that identity is the third check in Section Appendix A.3, below. The fraction-of-a-percent agreement reported in Section 3 at this study’s leads is the empirical statement that trajectory error stays small relative to Jacobian error here: most of the disagreement against MITGCM in Section 4.2 is $DF_\theta \neq DM$, not $x^F \neq x^M$.

Appendix A.2. The MITGCM/TAF reference adjoint

MITGCM’s adjoint is not approximated or hand-derived: it is produced by TAF (Transformation of Algorithms in Fortran, FastOpt), a source-to-source automatic-differentiation tool that differentiates the model’s own Fortran in reverse mode. The differentiation itself is a single step inside `slurm/mitgcm/af_s0_adjoint_build.sbatch: make adall` invokes `staf`, which submits the preprocessed forward source, `ad_input_code.f`, to FastOpt’s server and receives back the adjoint source, `ad_input_code_ad.f`. Every step before that is ordinary C-preprocessing and `genmake2` configuration; every step after it is an ordinary `gfortran` compile. `af_fno/mitgcm/input_ad/data.autodiff` requests the exact adjoint (`inAdExact = .TRUE.`), with no divided or approximate variant.

The control variable — the quantity the adjoint is taken with respect to — is the initial sea surface height, wired through `ALLOW_GENARR2D_CONTROL` in `af_fno/mitgcm/code_ad/CTRL_OPTIONS.h` and configured in `af_fno/mitgcm/input_ad/data.ctrl`. This checkout’s alternative, `ALLOW_ETANO_CONTROL`, compiles but is dead code — its implementation sits entirely inside a preprocessor guard that nothing in this build defines — so it silently returns a gradient of exactly zero; the live path is confirmed never to regress to it. The cost functional itself, $J = \sum_{ij} w_{ij} \eta_{ij}(T)$, is defined in `af_fno/mitgcm/code_ad/cost_test.F`, with the weight field w read as an ordinary constant, not a control, from a run-time `costWeight.bin`, wired in `af_fno/mitgcm/input_ad/data.cost`. The trajectory the adjoint is linearized about is staged by `scripts/stage_adjoint_run.py`, whose physics namelists are identical to the renderer that produced every forward training segment; only the run’s start iteration and dump frequency differ. The reverse sweep’s own memory management — how TAF

checkpoints the forward trajectory so it can be replayed backward without storing every timestep — is configured in `af_fno/mitgcm/code_ad/tamc.h` and is sized to cover the full window used in this report.

Trusting the resulting gradient rests on four independent checks, all run after the build and before any number in Section 4.2 is produced. First, a direct finite-difference test: the control is perturbed by $\pm\epsilon$ at a chosen cell, the forward model is run twice, and the centred difference is compared against the adjoint’s value at that cell, swept over $\epsilon \in \{10^{-3}, \dots, 10^{-6}\}$ so that agreement is judged by whether the error shrinks and then plateaus across four orders of magnitude rather than by a single favourable epsilon. This is configured in `af_fno/mitgcm/input_ad/data.grdchk` and run by `slurm/mitgcm/af_s0_adjoint_grdchk.sbatch`, which sweeps every target cell used in this study plus points along the western boundary current and one offshore point; the achievable precision is set by the pressure solver’s own convergence tolerance (`cg2dTargetResidual` = 10^{-7} , identical to every forward S0 segment), not by any looseness in the adjoint itself. Second, `scripts/extract_mitgcm_adjoint.py` checks internal consistency: the twenty-day run’s adjoint state at day 7210 must equal the independently produced ten-day run’s map; the returned sensitivity must be exactly zero on all 244 land cells and finite everywhere; and the cost value the model itself prints must equal an independently computed $\sum w_{ij}\eta_{ij}$ in NumPy. Third, the same script checks the one case whose answer is known analytically: with the cost set to the pure area-weighted mean of η , the returned sensitivity must equal the weight field itself at every lead, because this configuration’s implicit free surface with `exactConserv` conserves that mean exactly — a case with no free parameters to tune and no room for the two sides to agree by coincidence. Fourth, every downstream extraction records the SHA-256 of the executable that produced it, so a later run using a different build — a different TAF submission, a different compiler flag — is caught rather than silently mixed into the same dataset.

`tests/test_mitgcm_adjoint.py` encodes the compile-time and namelist half of all of this as ordinary tests — which preprocessor flags must be defined, which must not, the tile-local indexing that `af_s0_adjoint_grdchk.sbatch` depends on, the checkpoint budget in `tamc.h` — runnable without MITGCM, TAF or a GPU, so a change to the build configuration that would silently break the adjoint fails in continuous testing before it ever reaches a cluster job.

Appendix A.3. The FNO emulator’s reverse-mode gradient

The emulator side needs no separate adjoint code: `S_forced` and `S_free` are produced by ordinary reverse-mode automatic differentiation (PyTorch `autograd`) through the same trained network used for forecasting, implemented in `scripts/fno_adjoint_ft90.py`. `scripts/fno_adjoint_model.py` is a thin registry over that one script — which checkpoint, which report, which output directory — used identically for the frozen parent model and for every paired control/response replicate in this study, so the same differentiation machinery produces every map compared in this report.

Trust here starts with knowing that the thing being differentiated is exactly the thing that was deployed. `load_frozen_model` loads the published checkpoint with strict key

matching, asserts the loaded parameter count against the value recorded when the checkpoint was written, and freezes every weight; a companion check then walks every module confirming that no training-time hook, parametrization or spectral-norm estimator buffer is still attached to the loaded network, since any of those would mean the differentiated map is not the one the model actually deploys.

A genuine numerical defect was found and fixed here, and is verified rather than assumed away. The third-party spectral-convolution layer that holds 97.95% of this architecture’s parameters allocates its internal Fourier working buffer in single precision regardless of the model’s own dtype, so casting the model to double precision left almost the entire operator’s gradient computed in float32 while every downstream report claimed float64. `DoublePrecisionSpectralConv` in `scripts/fno_adjoint_ft90.py` patches that buffer to follow the model’s precision, and `verify_double_precision_spectrum` confirms the fix took effect twice over: by directly observing the buffer’s dtype during a forward pass, and by checking the adjoint identity $\langle v, Ju \rangle = \langle J^T v, u \rangle$ on the affected layer, which only holds to machine precision once the promotion is in place.

The Jacobian-vector products themselves are built inside `FrozenOperator` (`scripts/fno_adjoint_ft90.py`), which places the physical-to-normalized coordinate change and the wet-cell mask inside the autograd graph rather than applying them by hand outside it, so reverse-mode differentiation automatically carries every factor of the per-channel normalization scale and returns exactly zero on land — the same property MITGCM’s land check enforces, arrived at the same way, by construction rather than by post-hoc correction. `S_forced` is assembled by chaining one vector-Jacobian product per ten-day step backward through the MITGCM truth states (`forced_chain`); `S_free` by one forward pass along the emulator’s own rollout followed by one backward pass per objective (`free_chain`).

Four checks establish that this reverse-mode gradient is correct. A finite-difference sweep at the same eight cells and the same style of epsilon ladder as the MITGCM side, implemented in `scripts/fno_adjoint_ft90.py`, plays the same role as MITGCM’s own check, with a tighter tolerance justified by the absence of any solver-tolerance or convective-adjustment discontinuity on this side. A second, independent check computes the same directional derivatives by forward-mode automatic differentiation (`torch.func.jvp`, in `forward_versus_reverse`) — a completely different code path, accumulating in the opposite order, that involves no subtraction of nearly equal numbers and is therefore not limited by the cancellation a finite difference suffers from; agreement between the two modes to a measured machine-precision floor is stronger evidence than any finite difference alone can supply. A third check compares the two autoregressive chains directly: at lead 10, `S_forced` and `S_free` are the same computation by construction, and the two independent implementations are required to agree bit-for-bit, not merely to within tolerance. A fourth check runs the same analytically known case as the MITGCM side — the conserved basin mean — and reports, rather than grades, how far the emulator’s answer sits from the exact weight field at every lead; this is the check that surfaces the ≈ 0.38 -per-call damping of the one mode whose true answer is known, discussed in Section 4.2.a.

`tests/test_fno_adjoint.py` fixes the frozen contract this all depends on — channel indices, the day windows each experiment differentiates over, the checkpoint’s expected

parameter count and hash — as tests runnable without a GPU, the trained checkpoint, or MITGCM, so the same class of silent drift the MITGCM-side tests guard against is caught here too.

Appendix A.4. Keeping the comparison itself honest

Producing two independently verified sensitivity maps is not sufficient on its own; the two must also be asking the same question. This is enforced in code rather than left to convention: the comparison in `scripts/compare_adjoint_maps.py` refuses to run unless the MITGCM side’s own finite-difference check has already passed, and unless both sides’ cost-weight files hash identically and target the same grid cell — so a stale weight file or a silently moved target on either side fails the comparison outright rather than producing a plausible-looking but meaningless number. Both pipelines are required to share the same cost functional, the same three target cells, the same sign convention (a positive entry means raising sea level at the source raises the target), the same physical units, and the same treatment of land, and `scripts/adjoint_metrics.py` applies the four comparison metrics of Eqs. (10)–(13) identically to both maps over the same wet-cell mask. The one substantive modelling choice that makes the comparison meaningful rather than merely well-typed is that `S_forced` — the map actually compared to MITGCM throughout Section 4.2 — is linearized about the true MITGCM trajectory at every step, exactly as MITGCM’s own adjoint is; this is what lets a disagreement be read as Jacobian error rather than as the two sides having quietly drifted apart onto different trajectories.